Chapter 12

Interview Preparation

12.1 Introduction

Quantitative-finance interviews test more than isolated technical facts. A strong candidate must identify the type of institution and role, recognize the assessment format, structure an answer under time pressure, and connect analysis to a trading, investment, or risk decision. This chapter organizes preparation around those repeatable tasks.

Specific employer names and identifying details are intentionally omitted. Institution types are retained because they are useful predictors of interview emphasis. A proprietary trading or market-making firm will often emphasize probability, mental arithmetic, games, and market intuition; an investment bank may combine product knowledge, risk, coding, and behavioral assessment; an asset manager or institutional investor may place more weight on portfolio construction, research process, and communication; and a hedge fund or quantitative investment manager may emphasize modeling depth, statistics, and independent research judgment.

The most effective way to use this chapter is in three passes. First, learn the process maps and answer structures. Second, practice the cases and technical prompts without notes. Third, personalize every behavioral answer with evidence from projects, coursework, internships, or research.

12.2 The End-to-End Interview Process

The exact sequence varies, but most recruiting processes can be understood as a progression from broad screening to increasingly job-specific evidence. At each stage, the candidate should know three things: the format, the competency being tested, and the evidence needed to pass.

Stage	**What usually happens**	**What is being tested**	**How to prepare**
Role and application screen	Résumé review, eligibility checks, and sometimes a short application question.	Minimum technical fit, evidence of interest, and whether experience is relevant to the role.	Translate every major résumé bullet into a problem, method, result, and lesson. Prepare a role-specific explanation of why the work matters.

Stage	What usually happens	What is being tested	How to prepare
Recruiter or HR screen	A short call covering motivation, logistics, communication, and broad experience.	Clarity, genuine interest, basic role understanding, and absence of obvious fit concerns.	Prepare 60-second answers for "tell me about yourself," "why this role," and "why this institution type." Know dates, location constraints, and work authorization.
Online assessment	Timed mathematics, probability, coding, data analysis, pattern recognition, or numerical tests.	Speed, accuracy, fundamentals, and performance under constraints.	Rehearse in the actual time limit. Use a skip rule, track error types, and practice the platform mechanics before the scored attempt.
Pre-recorded video	Several behavioral or motivation questions with limited preparation and response time.	Structure, concision, presence, and consistency of motivation.	Practice 60–90 second answers on camera. Lead with the answer, use one example, and finish with relevance to the role.
First live interview	One or two interviews mixing behavioral questions with technical fundamentals.	Whether the candidate can reason interactively, accept hints, and explain work clearly.	Practice aloud with a partner. State assumptions, narrate the method, and check the result instead of silently calculating.
Take-home or case study	A coding assignment, research task, forecasting problem, investment case, or risk analysis, often followed by a presentation.	Problem definition, judgment, implementation quality, validation, and communication.	Define the target and evaluation metric first. Build a transparent baseline, document assumptions, test failure modes, and prepare a short executive summary.
Final round or superday	Several interviews with managers, technical specialists, and potential teammates.	Depth, consistency across interviewers, judgment, collaboration, and team fit.	Prepare for repeated questions without sounding scripted. Review every earlier submission and maintain the same assumptions, results, and motivation story.

Stage	What usually happens	What is being tested	How to prepare
Informal interaction	Lunch, coffee, office tour, or networking conversation.	Professionalism, curiosity, listening, and how the candidate behaves when the format feels less formal.	Ask thoughtful questions about work, feedback, and team decision-making. Treat the interaction as evaluated while remaining conversational.
Follow-up	Thank-you note, additional materials, references, or a clarification call.	Professional judgment and continued interest.	Send a concise note, correct any material error transparently, and avoid introducing new unsupported claims.

12.2.1 Typical Sequences by Institution Type

Institution type is often a better predictor of the process than the exact job title.

- **Proprietary trading or market-making firm:** timed mathematics or game assessment → probability and mental-arithmetic interview → trading games, market intuition, coding, and behavioral final rounds.
- **Hedge fund or quantitative investment manager:** mathematics or coding screen → statistics, machine learning, or research interview → take-home research task or deep project discussion → final interviews focused on judgment and research independence.
- **Investment bank or diversified financial institution:** recruiter or recorded-video screen → role-specific technical and behavioral interviews → multi-interviewer final round. Product knowledge, risk, controls, and communication may all appear.
- **Asset manager or institutional investor:** motivation and market discussion → investment, portfolio, or data case → presentation and panel interviews. The process often emphasizes research judgment and communication to stakeholders.
- **Insurance, risk, or advisory institution:** behavioral screen → statistics, financial-risk, product, or modeling interview → case study or manager panel. Governance and communication are often tested alongside technical knowledge.

12.2.2 How to Read Signals During the Process

The format itself tells the candidate how to allocate preparation time. A high-volume timed test rewards automaticity and a skip strategy. A take-home case rewards explicit assumptions, reproducibility, and documentation. A live technical interview rewards interaction: the interviewer needs to hear the reasoning and see how the candidate responds to hints. A final-round panel tests consistency; answers about motivation, projects, and technical choices should agree across interviewers.

12.3 A Repeatable Preparation Method

12.3.1 Step 1: Decode the Role

Reduce the job description to four columns: responsibilities, required knowledge, likely interview format, and personal evidence. Repeated verbs are especially informative. "Research" suggests hypothesis formation and validation; "build" suggests coding and systems design; "manage risk" suggests scenarios, limits, and communication; "trade" suggests probability, market intuition, and decisions under uncertainty.

12.3.2 Step 2: Build a Competency Map

For each role, score preparation needs from 0 to 3 across the following areas:

- probability, statistics, and mathematical reasoning;
- coding, algorithms, SQL, and data manipulation;
- machine learning, time series, and validation;
- derivatives, fixed income, portfolio theory, or product knowledge;
- market intuition and commercial judgment;
- behavioral evidence, communication, and motivation.

Spend most practice time on high-importance areas where current performance is weak. Do not distribute time evenly simply because the syllabus is broad.

12.3.3 Step 3: Build an Evidence Bank

Prepare six to eight reusable stories rather than a separate story for every behavioral question. The bank should cover leadership, conflict, failure, ambiguity, rapid learning, ethical judgment, analytical impact, and communication with a non-technical stakeholder. Each story should contain a measurable result and a sentence connecting it to the target role.

12.3.4 Step 4: Prepare Technical Answers at Two Depths

Every core topic should have a 30-second explanation and a 90-second explanation. The short version gives the definition, intuition, and one limitation. The longer version adds a formula or example, implementation detail, and follow-up. This prevents both one-word answers and uncontrolled lectures.

12.3.5 Step 5: Practice in the Real Format

Untimed study builds knowledge; format-matched rehearsal builds interview performance. Complete timed assessments under the real limit, record video answers with the real preparation window, code without autocomplete when appropriate, present cases from slides, and solve live technical questions while speaking.

12.3.6 Step 6: Keep an Error Log

After every practice session, classify misses as concept, setup, arithmetic, coding, time management, or communication errors. Record the correction and one new drill. Improvement should be measured by repeated performance, not by whether the solution looked familiar after reading it.

12.3.7 Preparation Deliverables

Before the first interview, the candidate should have the following compact set of materials:

- a one-page role and institution-type brief;
- a 60-second introduction and a 90-second motivation answer;
- six to eight behavioral stories;
- a project deep-dive sheet covering data, method, validation, result, limitations, and personal contribution;
- a technical formula and concept sheet tailored to the role;
- one timed-assessment strategy, including a skip rule;
- one mock case or coding exercise completed under realistic conditions;
- five thoughtful questions for interviewers at different seniority levels.

12.4 Role-Specific Process Examples

12.4.1 Commodities Trading (Summer Trading Analyst)

Typical institution type: Commodity merchant, energy trading firm, or commodities desk.

Observed interview format

- Case study focused on natural gas price prediction (features $\rightarrow$ next-year price).
- HR interview (trading interest, commodity preference, data experience).
- Superday: in-person, multiple rounds, including manager behavioral interview.
- Networking lunch (informal but evaluated).
- Brain teasers / estimation questions.
- Market intuition: event-driven scenario (e.g., refinery explosion).
- Technical: Python + pandas, stochastic process simulation, Monte Carlo, plus SQL (medium-difficulty practice problems).

12.4.2 Quantitative Risk Analyst

Typical institution type: Cross-industry quantitative-finance employers.

Core topics observed

- Value at Risk (VaR) and practical risk measurement.
- Data-driven risk modeling and stress testing.
- Delta–gamma option risk/trading questions.

12.4.3 Pre-Recorded Video Interview (pre-recorded video interview): Technology Audit Summer Analyst (Internal Audit)

Typical institution type: Investment bank or diversified financial institution.

Format observed

- Pre-recorded video interview (pre-recorded video interview).
- Questions were primarily behavioral and experience-based.
- Emphasis on communication clarity and structured examples.

Questions encountered

1. What interests you most about a career in Internal Audit?
2. How do you see the Summer Analyst Program contributing to your long-term career goals?
3. To what extent have you leveraged GenAI or other data analytics tools (Python, SQL, visualization and workflow-automation tools), and what impact will these technologies have on internal audit or financial services?

12.4.4 Pre-Recorded Video Interview (pre-recorded video interview): Investment Management Division (Portfolio Solutions Group)

Typical institution type: Asset management firm or institutional investment division.

Questions encountered

1. Why do you want to pursue a career in the Investment Management Division?
2. Which group(s) within Investment Management are you most interested in and why?
3. What is your greatest weakness in understanding finance/economics? What did you do to remediate it?
4. "Doing the right thing" is a core value. Describe a time you noticed something that did not appear right. How did you handle it?

12.4.5 Other Assessments Mentioned

Typical institution type: Multiple quantitative-finance institution types.

Macro hedge fund (Risk and Treasury Intern)

- Cognitive screening assessment (12 minutes).
- Python assessment (about 60 minutes).
- Group exercise: 3–4 people working on a stock model.

Other firms / platforms referenced

- Options market-making firm: online math assessment.
- Quant research: a timed mathematics assessment + cognitive section + brain teaser.
- Quantitative proprietary trading firm: online assessment.
- Electronic market-making firm: "Beat the Odds", number logic, and game-style tests (e.g., zap-N style tasks).

12.4.6 Quant Research Intern: Assessment Formats

Typical institution type: Hedge fund, quantitative investment manager, or proprietary trading firm.

Track A: Proctored Math Online Assessment

- Proctored online math assessment.
- Duration: 90 minutes.
- Volume: 16 questions.
- Topics: probability, expected value, combinatorics, brain teasers.

Track B: 24-Hour Take-Home Coding Assessment

- Take-home coding assessment with a 24-hour deadline.
- Content: object-oriented programming, understanding bid/ask spreads, arbitrage logic.
- Difficulty: conceptually easy, but long; completeness and code quality matter.

Track C: Proctored Python Coding (Mostly pandas)

- Proctored Python coding assessment focused on pandas.
- Emphasis: EDA, data wrangling, modeling, evaluation.
- Difficulty: generally easy, but sometimes requires highly specific pandas details (e.g., `int64` dtype).

12.4.7 Trading Track Online Assessments

Typical institution type: Proprietary trading or electronic market-making firm.

Track A: Long-Form Game-Theory Puzzle (Card Game Inference)

- A long puzzle session (about several hours).
- A card game was described with specific rules.
- Based on observed actions, the task was to infer which card(s) the person likely held.
- Required probability, expected value, and "optimal play" reasoning.

Track B: Cognitive Ability Test (Pattern + Memory Games)

- Puzzle / memory game format.
- Pattern recognition: choose the next geometric sequence.
- Working memory: click the order of dots that blinked (sequence recall).

Track C: Valuation and Expected Value Problems (EV/EBITDA + Default Risk)

- Finance/accounting flavored questions involving EV/EBITDA.
- Expected value problems for pricing assets under default probability or price decline scenarios.
- More straightforward calculations, but required comfort with accounting terms and valuation logic.

12.4.8 Trading Risk Management Intern

Typical institution type: Investment bank, market-making firm, hedge fund, or asset manager.

Observed interview format

- Total of 3 interviews.
- Round 1: basic HR phone screen (qualification check; no technical).
- Round 2: 30 minutes with 2 managers (mixed technical + behavioral).
- Round 3: 30 minutes with 2 managers (mixed technical + behavioral).

12.4.9 Quantitative Technology Intern and Related Roles

Typical institution type: Cross-industry quantitative-finance employers.

Track A: Quantitative Technology Intern (Three Interviewers)

- Round 1: Fixed Income Fundamentals.
- Round 2: NLP and Machine Learning.

Track B: Investment Analyst Intern (Endowment-Style Role)

- Preparation focus: portfolio construction, valuation, macro sensitivity, and communication.

Track C: Quantitative Advisory / Consulting Internship (Time-Series Validation)

- Theme mentioned: time-series validation during COVID regime.

Track D: Portfolio Management / Research Internship (Logistic Regression)

- Questions encountered: logistic regression, L1/L2 regularization, LASSO, and other regularization methods.

12.5 Behavioral and Fit Preparation

12.5.1 Trading and Commodities Roles

Typical institution type: Commodity merchant, energy trading firm, or commodities desk.

1) "What experience do you have with trading?"

- Use a 60-second narrative: motivation → what you actually did → what you learned.
- Include at least one concrete example: a backtest, a simulated strategy, paper trading, or risk analysis.
- Close with relevance: "This is why I want a trading analyst role and why I will be effective quickly."

2) "Which commodity are you interested in?"

- Pick one and give a structured reason:
  1. Market structure (seasonality, supply chain constraints, multiple drivers).
  2. Data availability (public reports, inventories, weather).
  3. What signals you would explore and why they might exist.
- Avoid vague answers like "it is interesting"; state 2–3 drivers and a plausible mechanism.

3) "What data have you encountered?"

- Answer with type + scale + hard part + how you fixed it.
- Examples: time series (prices, volumes), alternative data (text/news), panel data, event data.
- Mention one reliability practice: unit tests, data validation checks, reproducible pipeline.

In-person superday and networking lunch

How to prepare

- Have 2–3 short stories ready: teamwork, disagreement, fast learning, handling ambiguity.
- During lunch: be professional, curious, and consistent; avoid controversial topics.
- Ask good questions: how they evaluate signals, what makes interns successful, what the team values.

12.5.2 Technology Audit Summer Analyst (Internal Audit)

Typical institution type: Investment bank or diversified financial institution.

1. Internal Audit motivation (60–90 seconds)

- Start with a one-sentence thesis: "I like work that combines systems thinking, controls, and measurable impact."
- Name 2–3 audit themes: risk identification, control design/testing, stakeholder communication, continuous monitoring.
- Give one concrete example from a project where you found/mitigated risk, improved data quality, or built monitoring.
- Close with why audit is a fit: breadth across businesses, disciplined process, and learning curve.

2. Program → long-term goals

- Goal: become strong in risk/control frameworks and technical auditing (data, systems, process).
- Program contributions: rotation exposure, mentorship, real deliverables, cross-functional partners.
- Anchor with a specific growth plan: "I want to strengthen X (controls testing, data lineage, analytics) by doing Y (audits, dashboards, automation)."

3. GenAI / analytics tools question

- Inventory your tools: 1–2 lines each for Python/SQL/BI tools and what you used them for.
- Impact: quantify: reduced manual checks, faster reconciliation, automated anomaly detection.
- Future impact: continuous auditing, smarter sampling, narrative reporting, but note governance: model risk, privacy, access control, prompt/data leakage, reproducibility.
- Safeguards: human-in-the-loop, audit trails, versioned code, documented assumptions.

12.5.3 Investment Management Division (Portfolio Solutions Group)

Typical institution type: Asset management firm or institutional investment division.

1. Why investment management

- Start with interest: decision-making under uncertainty, long-horizon thinking, portfolio construction.
- Show you understand the job: research, risk budgeting, implementation, client constraints.
- Provide one example: building a strategy, evaluating factors, measuring risk/return.

2. Which group and why

- Mention 1–2 groups and tie to specific work outputs (asset allocation, factor exposure control, client portfolio solutions).
- Explain your comparative advantage: data, modeling, communication with stakeholders.
- Ask/answer: "What problems does this group solve? What metrics define success?"

3. Weakness + remediation

- Choose a real but non-fatal weakness (e.g., macro intuition, accounting details, derivatives mechanics).
- Show a plan: textbooks/courses, structured reading, building a project/backtest, discussing with peers.
- End with outcome: what you can do now that you could not before.

4. Ethics / "doing the right thing"

- Use STAR: Situation, Task, Action, Result.
- Emphasize escalation path: clarify facts, consult policy/mentor, document, escalate appropriately.
- Show balanced judgment: protect clients/firm, avoid assumptions, resolve respectfully.

12.5.4 Trading Risk Management Intern

Typical institution type: Investment bank, market-making firm, hedge fund, or asset manager.

1) "How do you deal with slackers in a group project?"

- Start with diagnosis: confirm expectations, responsibilities, and constraints (sometimes the issue is unclear scope).
- Private alignment first: speak 1:1 to understand blockers; avoid public blaming.
- Make work visible: break tasks into small deliverables with dates; use a shared tracker.
- Rebalance if needed: reassign tasks to match strengths; define a minimal viable contribution.
- Escalate appropriately: if repeated non-delivery, escalate to the project lead with facts and proposed solution.
- Close with outcome: emphasize learning and maintaining team morale while protecting deadlines.

2) "Why did you pick your master's program?"

- Use a 3-part structure: goal (skills you wanted), fit (courses/projects), evidence (what you produced).
- Tie to the role: time series, risk, derivatives, data pipelines, or modeling.

3) "Why this role / why this firm?"

- Show understanding of the day-to-day: risk measurement, limit monitoring, stress/scenario work, desk communication.
- Mention 1–2 specific learning goals (rates products, swaps, curve risk, model governance).
- Connect to past experience: similar risk project, fixed income coursework, or data/automation work.

4) Resume questions

- Prepare 2–3 "deep dives" you can explain: problem, data, method, validation, impact.
- Anticipate follow-ups: assumptions, failure cases, and what you would improve.

12.5.5 Quantitative Technology Intern: Motivation and Prior Experience

Typical institution type: Cross-industry quantitative-finance employers.

Questions encountered

- Why are you interested in this position?
- Prior banking experience (discussion and impact).

How to answer (recommended structure)

- Use a 60–90 second narrative: why the role, why the team, and how your skills map to deliverables.
- Prior experience: pick one project and give Problem → Method → Result (quantified) → Lesson.

12.6 Case Studies, Estimation, and Market Intuition

12.6.1 Case Study: Natural Gas Price Prediction

Typical institution type: Commodity merchant, energy trading firm, or commodities desk.

What they are testing

- Whether you can translate a business question into a forecasting/feature problem.
- Whether you can reason about drivers (fundamentals) and also validate with data.
- Whether you avoid common time-series mistakes (leakage, look-ahead bias, wrong validation split).
- Whether you can communicate trade-offs: interpretability vs accuracy, simplicity vs complexity.

A strong answer structure (use this template)

1. **Clarify target and horizon.** Define what "price next year" means: average annual price, year-end spot, front-month futures in 12 months, etc. Confirm frequency (daily/weekly/monthly) and whether predicting level or return.

2. **Build a fundamental map of drivers.** Before touching code, list plausible driver categories:

   - Supply: production, rig count, pipeline constraints, LNG exports.
   - Demand: weather (HDD/CDD), industrial usage, power burn.
   - Storage/inventory: EIA storage levels, seasonal patterns.
   - Macro/energy complex: crude prices, coal substitution, rate environment.

   Then match dataset columns to these categories and explain why each may lead/lag price.

3. **Data checks and feature engineering.** Explain your steps:

   - Missingness/outliers: check missing patterns; winsorize extreme errors; do not impute future information.
   - Stationarity: consider modeling returns/log changes; seasonality features (month-of-year).
   - Lag features: use only information available at time t to predict $t + \Delta$.
   - Leakage audit: confirm no post-horizon features sneak in (e.g., future realized inventory).

4. **Baseline model first.** Start with a simple benchmark:

   - Naive: last value / seasonal average.
   - Linear regression with lags and seasonal dummies.
   - Ridge/LASSO for feature selection.

   Explain that beating the baseline is the key, and complexity comes later.

5. **Validation (time series).** Use rolling/expanding window:

   - Train on early period $\rightarrow$ validate on later period, roll forward.
   - Evaluate MAE/MAPE for levels; directional accuracy if relevant.
   - Report stability across regimes (high vol vs low vol periods).

6. **Interpret results.** Use:

   - Feature importance (linear coefficients, permutation importance, SHAP).
   - Sensitivity analysis: how prediction changes if storage increases, temperature shocks, etc.
   - Error analysis by season/regime.

7. **Deliver the trading interpretation.** Forecast $\rightarrow$ how to trade or risk-manage: direction, conviction, risk limits, scenario ranges, and what would falsify the signal.

Preparation checklist for this case study

- Understand basic natural gas market drivers (storage seasonality, weather, LNG export sensitivity).
- Be able to describe a time-series CV approach without leakage.

- Practice: build a small forecasting notebook on any commodity time series with lag features and rolling validation.
- Prepare "model upgrade" options: tree models, regime features, Bayesian shrinkage; justify only if baseline is solid.

12.6.2 Brain Teasers / Estimation Questions

Typical institution type: Cross-industry quantitative-finance employers.

> Practice the same topic at two levels: role-specific application questions and general mathematical-foundation questions.

Common prompts

- "Estimate how many people are flying on airplanes right now in the world."
- "How many tennis balls fill this room?"

What they are testing Structure, assumptions, arithmetic discipline, and ability to sanity-check.

A reusable framework (Fermi template)

1. Define what exactly is being counted (snapshot vs per day, global vs region).
2. Break the problem into multiplicative components.
3. Choose reasonable assumptions (say them out loud).
4. Compute with clean units and intermediate checks.
5. Provide a range and sensitivity: "If flights/day is 80k vs 120k, the answer scales linearly."

Example: people flying right now (outline only) Flights per day $\times$ average flight duration / 24 hours $\times$ passengers per flight. State assumptions, compute a range, compare to known rough references (global daily passengers).

12.6.3 Market Intuition: "A refinery exploded. What happens?"

Typical institution type: Commodity merchant, energy trading firm, or commodities desk.

How to answer

1. Identify the product impacted (gasoline, diesel, jet fuel) and the regional market.
2. Translate to supply shock: immediate reduction in refinery throughput.
3. Discuss spreads: crack spreads widen; spot differentials jump; inventory draws.
4. Substitution and logistics: imports, pipeline reroutes, blending changes.
5. Time scale: immediate spike, then mean reversion as supply adjusts (unless outage is long).

12.7 Coding, Data, and Modeling

12.7.1 Python / pandas Expectations

Typical institution type: Cross-industry quantitative-finance employers.

- Data loading, cleaning, joins/merge, `groupby`, rolling windows, resampling.
- Avoid slow loops; use vectorization; explain time complexity if asked.

12.7.2 Stochastic Process Simulation (How to Present)

Typical institution type: Cross-industry quantitative-finance employers.

Practice the same topic at two levels: role-specific application questions and general mathematical-foundation questions.

1. Define the model first: for example GBM, OU, AR(1), or GARCH, and say clearly whether it is continuous-time or discrete-time.
2. State the model-specific evolution rule rather than forcing every process into one generic formula. For example, GBM is commonly written as $S_{t+\Delta} = S_t + \mu S_t \Delta + \sigma S_t \sqrt{\Delta}\, Z$, while OU uses mean reversion $X_{t+\Delta} = X_t + \kappa(\theta - X_t)\Delta + \sigma\sqrt{\Delta}\, Z$.
3. Generate paths, compute summary statistics, and validate them against what the model implies (mean, variance, stationarity, or reversion speed as appropriate).

A good interview answer sounds model-aware. Saying "I would discretize the process" is not enough; name the drift, diffusion, or discrete update that belongs to the specific process you chose.

12.7.3 Monte Carlo Solution Template

Typical institution type: Cross-industry quantitative-finance employers.

Practice the same topic at two levels: role-specific application questions and general mathematical-foundation questions.

1. Define random variables and payoff metric.
2. Simulate N paths; compute estimator $\hat{\theta}$.
3. Report standard error $SE_c \approx s/\sqrt{N}$ and confidence interval.
4. Add variance reduction if time allows (antithetic/control variates).

12.7.4 SQL Practice Targets (medium-difficulty practice problems)

Typical institution type: Cross-industry quantitative-finance employers.

- Joins, window functions (`ROW NUMBER`, `LAG`, `SUM OVER`), grouped aggregations, conditional logic.
- Practice writing queries with clear logic and minimal subqueries.

12.7.5 Missing Data or Alternative Data: What to Do?

Typical institution type: Cross-industry quantitative-finance employers.

1. Diagnose missingness (MCAR/MAR/MNAR) and create missing indicators.
2. Choose imputation method appropriate to data type and leakage risk: median/quantile, forward-fill with caps (time series), KNN/MICE, model-native missing handling (trees).
3. Validate inside CV folds; stress test sensitivity to imputation choices.
4. For alternative data: document provenance, stability, and potential bias; monitor drift and revisions.

12.7.6 NLP and Machine Learning Interview Questions

Typical institution type: Cross-industry quantitative-finance employers.

Questions encountered

- How do you deal with missing data in your project? (median imputation)
- Why not use the mean/average?
- Label encoding vs one-hot encoding: what is the fundamental difference and how do you apply them?
- Walk me through your ML experience on your resume.
- How did you do model validation?

How to answer (step-by-step)

1. **Missing data.**
   - First classify missingness: MCAR/MAR/MNAR; add missingness indicators if predictive.
   - Median vs mean: median is robust to outliers/skew; mean can be distorted by heavy tails.
   - Do imputation within CV folds to prevent leakage.
2. **Encoding.**
   - Label encoding assigns integers and may impose fake ordering; OK for ordinal variables and many tree models.
   - One-hot avoids ordering, better for linear models/NNs; costs dimensionality and sparsity.
   - High-cardinality categories: target encoding with K-fold regularization or embeddings.
3. **Model validation (generic).**
   - Split design: IID vs time series (rolling/expanding windows for time series).
   - Metrics: choose aligned with task (AUC/logloss for classification; RMSE/MAE for regression).

- Calibration: reliability plots; threshold selection; error analysis by segment.
- Robustness: stress periods, ablations, and sensitivity checks.

4. **Resume ML walkthrough.**
   - Problem statement → data → feature engineering → model choice → validation → impact.
   - Mention at least one engineering element: pipeline, reproducibility, tests, monitoring.

12.7.7 Time-Series Validation During COVID Regime

Typical institution type: Cross-industry quantitative-finance employers.

How to answer (time-series CV best practices)

1. Use rolling/expanding-window splits; never shuffle time.
2. Prevent leakage for overlapping labels: purged splits + embargo.
3. Evaluate performance by regime: pre-COVID vs COVID vs post-COVID; include stability metrics.
4. Document assumptions and add monitoring for drift (PSI, residual tests, volatility changes).

12.7.8 Portfolio Management / Research Internship: Logistic Regression

Typical institution type: Asset manager or institutional investment firm.

Questions encountered

- How to build logistic regression: inputs, outputs, difficulties.
- Did you use L1 and L2? What is LASSO?
- Did you use other regularization methods?

How to answer (step-by-step)

1. **Model definition.** Binary target $y \in \{0, 1\}$; features $x \in \mathbb{R}^d$.

$$p(y = 1 \mid x) = \sigma(\beta_0 + \beta^\top x), \qquad \sigma(u) = \frac{1}{1 + e^{-u}}.$$

2. **Training objective.** Negative log-likelihood (cross-entropy):

$$\mathcal{L}(\beta) = \sum_i \left[\log(1 + e^{\beta^\top x_i}) - y_i(\beta^\top x_i) \right].$$

3. **Difficulties.**
   - Class imbalance: use class weights, threshold tuning, PR-AUC.
   - Multicollinearity: regularize; feature scaling.
   - Complete separation: L2/Bayesian priors.
   - Non-stationarity: rolling re-fit, regime validation.

4. **Regularization.**
   - L2 (Ridge): $\mathcal{L} + \lambda\|\beta\|_2^2$; stabilizes estimates.
   - L1 (LASSO): $\mathcal{L} + \lambda\|\beta\|_1$; induces sparsity.
   - Elastic Net: combination for correlated features.
5. **Validation.** Nested CV; calibration; out-of-time tests; ablation and sensitivity.

12.8 Risk, Fixed Income, Options, and Valuation

Practice the same topic at two levels: role-specific application questions and general mathematical-foundation questions.

12.8.1 Value at Risk and Risk Modeling

Typical institution type: Investment bank, market-making firm, hedge fund, or asset manager.

VaR: a high-quality answer in 60–90 seconds

1. Define VaR precisely (confidence α, horizon H, loss quantile).
2. Give 3 methods: historical, parametric (delta-normal), Monte Carlo.
3. Mention backtesting and limitations (tail risk, assumptions, liquidity).

Risk modeling: what to propose

- Factor model (PCA/fundamental factors) for covariance stability.
- Heavy tails (Student-t), volatility clustering (GARCH/EWMA).
- Scenario analysis and stress testing; explain model governance and monitoring.

12.8.2 Stress Testing Using Data Methods

Typical institution type: Investment bank, market-making firm, hedge fund, or asset manager.

1. Choose scenarios: historical replay + hypothetical shocks.
2. Map shocks to risk factors (rates, spreads, vol, FX) and revalue positions.
3. Add liquidity and closeout assumptions.
4. Report: loss distribution, top contributors, and limit breaches.

12.8.3 Delta–Gamma Questions

Typical institution type: Investment bank, market-making firm, hedge fund, or asset manager.

- Use the second-order approximation:

$$\Delta V \approx \Delta S \cdot \Delta + \frac{1}{2}(\Delta S)^2 \Gamma.$$

- Explain: delta hedging removes first-order risk; gamma remains and drives curvature.
- Discuss gamma hedging with an additional option; mention vega/theta trade-offs.

12.8.4 Trading Risk Management: Technical Questions and How to Answer

Typical institution type: Investment bank, market-making firm, hedge fund, or asset manager.

1) Define Value at Risk (VaR).

1. Specify horizon H and confidence α.
2. Define VaR as a quantile of the loss distribution:

$$\mathrm{VaR}_\alpha(H) = \inf\{\ell : \mathbb{P}(\mathrm{Loss}_H \le \ell) \ge \alpha\}.$$

3. Mention 3 standard computation methods: historical simulation; parametric (delta-normal); Monte Carlo (full revaluation).
4. Limitations (one sentence): VaR does not describe tail severity; ES is more informative for tails.

2) Pricing mechanism of swaps (explain).

1. A vanilla interest rate swap exchanges fixed vs floating cash flows on a notional N.
2. The par swap rate S is chosen so that the swap has zero value at inception:

$$\mathrm{PV}(\text{fixed leg}) = \mathrm{PV}(\text{float leg}).$$

3. With discount factors $D(t_i)$ and accrual factors Δ_i,

$$\mathrm{PV}(\text{fixed}) = NS\sum_{i=1}^{n}\Delta_i D(t_i).$$

   For a standard par swap, the floating leg PV at inception is approximately

$$\mathrm{PV}(\text{float}) \approx N\big(1 - D(t_n)\big),$$

   hence

$$S = \frac{1 - D(t_n)}{\sum_{i=1}^{n}\Delta_i D(t_i)}.$$

4. Mention curve bootstrapping (OIS for discounting, IBOR or SOFR-based curves for projection) if asked.

3) Compute the present value of a bond on the spot (mental math).

1. Write cash flows: coupons C at $t_1, \ldots, t_n$ plus principal F at T.
2. Discount using yield/spot curve:

$$P = \sum_{i=1}^{n} C\,D(t_i) + F\,D(T), \qquad D(t) = e^{-z(t)t}.$$

3. Mental math tips: approximate discount factors for small rates, $e^{-rt} \approx 1 - rt$; group cash flows; check price bounds: if coupon $>$ yield, price $>$ par; if coupon $<$ yield, price $<$ par.

4) "Why are interest rates more volatile as time passes?"

- Clarify which quantity: short rate volatility vs long-maturity yield volatility vs bond price volatility.
- A standard interpretation: as horizon increases, uncertainty accumulates. For a random walk-like factor, variance grows with time (e.g., $\mathrm{Var}(W_t) = t$), so longer horizons can exhibit larger uncertainty in integrated quantities.
- For bonds, price sensitivity to rate changes is driven by duration: longer maturity generally has higher duration, hence larger price moves for a given yield shock.
- Long yields can be smoother if they average expected future short rates; in practice, term premium and regime changes still create meaningful long-horizon variance.

5) Statistics question about distributions (generic approach).

- Identify distribution family and parameters; compute mean/variance; translate the question into a probability or expectation.
- Use conditioning, CLT, or tail approximations if exact forms are hard.
- Provide sanity checks (limits, symmetry, scaling).

12.8.5 Fixed Income Fundamentals

Typical institution type: Investment bank, asset manager, insurer, or institutional investor.

1. What is a bond?

- Definition: a security that pays scheduled cash flows (coupons + principal).
- Pricing: present value of cash flows discounted by the term structure.
- Risks: rate risk, credit risk, liquidity, embedded options (call/prepay).

2. What is duration?

- Macaulay duration = PV-weighted average time of cash flows.
- Modified duration approximates first-order price sensitivity: $\Delta P/P \approx -D_{\mathrm{mod}}\,\Delta y$.
- Mention convexity as second-order correction.

3. How to calculate duration (step-by-step).

1. Build cash flow schedule $\{(t_i, CF_i)\}$.
2. Discount each cash flow using spot rates $z(t_i)$ or a yield y: $PV_i = CF_i e^{-z(t_i)t_i}$.
3. Compute price $P = \sum_i PV_i$.
4. Macaulay duration $D_{\mathrm{Mac}} = \dfrac{1}{P}\sum_i t_i PV_i$.

5. Modified duration (consistent compounding convention): $D_{\text{mod}} = D_{\text{Mac}}/(1+y)$.
6. For curve risk: compute key-rate durations by bumping one tenor at a time.

4. Yield curve (what to say when asked "tell me more").

- Define: mapping maturity t to spot/zero rates $z(t)$ or discount factors $D(t)$.
- Construction: bootstrap from liquid instruments (bills/OIS/swaps); interpolate smoothly.
- Risk: interpret moves via level/slope/curvature; manage via key-rate DV01.
- Use cases: pricing, hedging, forward rates, scenario/stress testing.

12.8.6 Investment Analyst Intern (Endowment-Style Role)

Typical institution type: Endowment, foundation, outsourced investment office, or institutional investor.

Preparation focus (typical)

- Portfolio construction: diversification, factor exposures, drawdown and risk budgeting.
- Valuation: basic multiples, cash flow logic, understanding macro sensitivity.
- Communication: ability to summarize an investment thesis concisely and defend assumptions.

12.8.7 Valuation and Expected Value Problems (EV/EBITDA + Default Risk)

Typical institution type: Investment bank, asset manager, insurer, or institutional investor.

How to answer these questions (step-by-step)

1. Define the payoff clearly. Identify states of the world (default / no default; price up/down), and write payoff X in each state.
2. Compute fair price as discounted expectation. For horizon T with discount rate r:
$$P_0 = e^{-rT}\,\mathbb{E}[X] = e^{-rT}\sum_s p_s X_s.$$
3. Include recovery and loss-given-default when needed. If default probability is p and recovery is R (fraction of face value F):
$$\mathbb{E}[X] = (1-p)\cdot F + p\cdot(RF).$$
4. Connect EV/EBITDA to enterprise value. Recall definitions:
$$EV = \text{Market Cap} + \text{Debt} - \text{Cash},$$
and the multiple EV/EBITDA is used to infer EV from EBITDA:
$$EV = (EV/EBITDA)_{\text{peer}} \times EBITDA.$$
Then back out equity value via Equity $= EV - \text{Debt} + \text{Cash}$.
5. Sanity checks. Make sure units match, probabilities sum to 1, and price lies between extreme outcomes (after discounting).

12.9 Assessment Formats and Strategy

12.9.1 Preparation Focus Across Mentioned Assessment Formats

Typical institution type: Multiple quantitative-finance institution types.

- Time pressure: practice doing clean reasoning quickly and writing concise solutions.
- Coding: Python data structures, complexity, vectorization, and debugging speed.
- Math: probability distributions, calculus (derivatives/integrals), linear algebra basics.
- Communication: in group exercises, narrate assumptions, assign roles, and keep the team aligned.

12.9.2 Proctored Math Online Assessment (Quant Research Intern)

Typical institution type: Hedge fund, quantitative investment manager, or proprietary trading firm.

> For timed breadth, draw heavily from a suitable practice set: calculus/linear algebra, probability, Markov chains, martingales, and Brownian motion. Keep the easier questions to a 3–5 minute target and reserve longer dynamic-programming questions for the end.

What this format is testing

- Speed and accuracy under time pressure.
- Breadth across probability/combinatorics (not just "light" mental math).
- Ability to pick the right technique quickly (symmetry, conditioning, recursion, linearity of expectation).

How to approach each question (repeatable template)

1. Classify the question in 5–10 seconds: EV? counting? conditional probability? recursion? geometry?
2. Write the key random variable(s). State exactly what you are computing (e.g., $\mathbb{E}[X]$, $\mathbb{P}(A \mid B)$).
3. Look for shortcuts first.
   - Linearity of expectation even without independence.
   - Symmetry (exchangeable outcomes).
   - Complement (compute $1 - \mathbb{P}(\cdot)$ if easier).
   - Condition on the first step for processes.
   - Indicator variables for counts.
4. Compute cleanly. Keep fractions exact; postpone decimals.
5. Sanity-check. Is the answer in $[0, 1]$? Does it match edge cases?

Preparation plan (90 minutes / 16 questions)

- Target pacing: ∼5 minutes/question average; plan 10–15 minutes buffer for 2–3 hard ones.
- Drill sets: binomial/negative binomial, geometric, hypergeometric; coupon collector variants; Markov chains with 2–3 states; classical combinatorics.
- Brain teasers: practice Fermi estimation, invariants, and "constructive" logic.
- Build an error log of common misses: off-by-one, conditional sample space, double counting.

12.9.3 24-Hour Take-Home Coding Assessment (Quant Research Intern)

Typical institution type: Hedge fund, quantitative investment manager, or proprietary trading firm.

How to structure your solution (what reviewers look for)

1. Restate requirements. Write a brief spec: inputs, outputs, constraints, edge cases.
2. Design first. Identify the core abstractions and data flow.
   - Quote (bid, ask, timestamp), Order, Trade, Venue, Book.
   - A pricing/arb module: detect violations like bid > ask across venues or implied cross-rate arbitrage.
3. Implement incrementally. Start with minimal features + tests; then add enhancements.
4. Correctness first. Unit tests for each module; golden examples.
5. Performance and clarity. Avoid premature micro-optimizations, but use appropriate data structures (dict/heap/Deque).
6. Documentation. README: how to run, assumptions, complexity.

Spreads and arbitrage: what to explain clearly

- Spread = ask – bid; midpoint = (ask + bid)/2; crossing markets if bid ≥ ask.
- Cross-venue arb: if venue A bid > venue B ask, buy at B ask, sell at A bid (after costs/latency).
- Include frictions: fees, slippage, minimum trade size, stale quotes; otherwise "arb" is illusory.

Take-home time management

- First 1 hour: plan + write skeleton + tests.
- Next 3–6 hours: implement core logic.
- Last 1–2 hours: refactor, add edge cases, improve README, run lint/tests.

12.9.4 Proctored Python Coding (Mostly pandas)

Typical institution type: Cross-industry quantitative-finance employers.

How to answer efficiently (pandas assessment playbook)

1. Understand the schema. `df.info()`, `df.head()`, missingness and unique counts.
2. Clean deterministically.

- Parse dates with `pd.to_datetime`.
- Enforce dtypes: `astype('int64')` / nullable `'Int64'` when missing values exist.
- Handle missing values explicitly and consistently.

3. Wrangling patterns to know.
   - `merge/join`, `groupby().agg()`, `pivot_table`, `sort_values`.
   - Window operations: `rolling`, `expanding`, `shift`, `diff`.
4. Baseline model first. Train/test split (or time-based split), then logistic/linear model baseline.
5. Evaluation. Choose metric aligned with task: RMSE/MAE for regression; AUC/F1/logloss for classification.
6. Edge-case checks. Duplicate keys, leakage, label imbalance, unexpected categories.

Common pitfalls to avoid

- Confusing `int64` vs pandas nullable `Int64`; failing type checks in hidden tests.
- Silent merges that increase row count due to many-to-many keys.
- Leakage (using target-derived features or future information).
- Using chained assignment or not handling NaNs before casting.

A minimal "dtype-safe" template (snippet)

```
import pandas as pd

def coerce_int64(series: pd.Series) -> pd.Series:
    # Use pandas nullable integer if NaNs exist
    if series.isna().any():
        return series.astype("Int64")
    return series.astype("int64")
```

12.9.5 Long-Form Game-Theory Puzzle (Card Game Inference)

Typical institution type: Proprietary trading or electronic market-making firm.

What this format is testing

- Strategic reasoning under uncertainty (game-theoretic optimality).
- Bayesian-style inference: actions provide information about hidden states.
- Ability to write detailed justification: not just the answer, but why it follows from optimal play.
- Consistency: your beliefs and your recommended strategy should align.

How to solve these problems (recommended template)

1. Restate the game formally.

- Define the hidden state (the card type or hand).
- Define the action set (what a player can do at each step).
- Define payoffs and stopping conditions.
- Identify what information each player has (information sets).

2. Write down the optimality principle.
   - If the other player is rational, their action should maximize expected payoff given their information.
   - Use backward induction if the game has a finite horizon; otherwise use threshold logic or value functions.
3. Connect actions to beliefs (inference).
   - For each possible card/hand, compute what the rational action would be.
   - Eliminate states inconsistent with the observed action.
   - If multiple states remain, assign probabilities using Bayes' rule:

$$\mathbb{P}(H \mid A) = \frac{\mathbb{P}(A \mid H)\mathbb{P}(H)}{\sum_h \mathbb{P}(A \mid h)\mathbb{P}(h)}.$$

4. Compute expected values and compare actions.
   - For each action, compute EV under your posterior beliefs.
   - Pick the action with the highest EV (or highest utility if risk-averse utility is specified).
   - Provide a sensitivity check: "If the prior were different, the threshold would move."
5. Write the justification like a proof.
   - State assumptions (rationality, priors, tie-breaking).
   - Show each elimination step or EV comparison clearly.
   - End with a concise conclusion: posterior odds and the recommended action.

How to prepare

- Practice expected value games: optimal stopping, mixed strategies, threshold policies.
- Practice inference from actions: signaling games, "rational agent" puzzles.
- Practice writing: for each puzzle, force yourself to write a 5–10 sentence justification.
- Time management for long sessions: take structured notes; freeze definitions; avoid re-reading rules repeatedly.

12.9.6 Cognitive Ability Test (Pattern + Memory Games)

Typical institution type: Proprietary trading or electronic market-making firm.

What they are testing

- Speed-accuracy tradeoff under strict time limits.
- Working memory capacity and attention control.
- Visual reasoning: transformations (rotation, reflection, scaling) and rule extraction.

How to perform better (practical tips)

1. Pattern tasks: scan for invariants first (counts, symmetry, parity), then transformations (rotate/flip/shift), then arithmetic rules (increment, alternate).
2. Memory tasks: chunk sequences (e.g., 2-2-3), map to a grid coordinate system, and keep a consistent reading order.
3. Pacing: do not get stuck. If uncertain, guess quickly and move on; the scoring is often throughput-sensitive.
4. Practice: short daily drills outperform occasional long sessions.

12.9.7 Valuation and Expected Value Problems in Online Assessments

Typical institution type: Investment bank, asset manager, insurer, or institutional investor.

How to prepare

- Drill expected value pricing with discrete scenarios (default/recovery, barrier events, partial payouts).
- Memorize core valuation identities (EV, equity value bridge).
- Practice quick accounting interpretations: what changes EV vs EBITDA, how leverage affects equity.

12.10 Selected Technical Practice Questions

The seven worked examples below stay compact on purpose. After each example, add one calculus, linear-algebra, or probability variation.

12.10.1 Macro Hedge Fund — Technical Questions (with solutions)

Typical institution type: Macro hedge fund.

Q1. Differential equations (basic)

Problem. Solve the linear ODE $y'(x) = ay(x) + b$ with $y(0) = y_0$.

Solution. Rewrite $y' - ay = b$. Multiply by integrating factor e^{-ax}:

$$\frac{d}{dx}\big(e^{-ax}y\big) = be^{-ax}.$$

Integrate: $e^{-ax}y - y_0 = \dfrac{b}{-a}(e^{-ax} - 1)$ if $a \neq 0$. Thus

$$y(x) = \left(y_0 + \frac{b}{a}\right) e^{ax} - \frac{b}{a} \qquad (a \neq 0), \qquad y(x) = y_0 + bx \qquad (a = 0).$$

This is a standard integrating-factor problem. The safest way to avoid sign mistakes in an interview is to check the initial condition immediately at $x = 0$ before you move on.

Q2. Two random points on a circle: chord longer than the radius

Problem. Pick two points uniformly at random on a circle of radius R. What is the probability the chord length exceeds R?

Solution. Let $\theta \in [0, \pi]$ be the smaller central angle between the points. For uniform points, θ is uniform on $[0, \pi]$. Chord length is $L = 2R\sin(\theta/2)$. We want $L > R$:

$$2R\sin(\theta/2) > R \iff \sin(\theta/2) > \frac{1}{2} \iff \theta/2 > \frac{\pi}{6} \iff \theta > \frac{\pi}{3}.$$

Therefore

$$\mathbb{P}(L > R) = \frac{\pi - \pi/3}{\pi} = \frac{2}{3}.$$

Q3. Differentiate an integral with variable limit (Fundamental Theorem)

Problem. If $F(x) = \int_0^x s\,ds$, compute $F'(x)$.

Solution. By the Fundamental Theorem of Calculus, $F'(x) = x$. (Also $F(x) = \frac{1}{2}x^2$ so derivative is x.) Hence $F'(x) = x$.

Q4. Four fair dice: probability the maximum equals 3

Problem. Roll 4 fair six-sided dice. What is $\mathbb{P}(\max = 3)$?

Solution.

$$\mathbb{P}(\max \leq 3) = \left(\frac{3}{6}\right)^4 = \left(\frac{1}{2}\right)^4 = \frac{1}{16}, \qquad \mathbb{P}(\max \leq 2) = \left(\frac{2}{6}\right)^4 = \left(\frac{1}{3}\right)^4 = \frac{1}{81}.$$

Thus

$$\mathbb{P}(\max = 3) = \mathbb{P}(\max \leq 3) - \mathbb{P}(\max \leq 2) = \frac{1}{16} - \frac{1}{81} = \frac{65}{1296}.$$

Q5. Orthogonal matrix characterization

Problem. For a matrix A, what does it mean for A to be orthogonal?

Solution. A is orthogonal iff it preserves dot products and lengths:

$$A^\top A = I \qquad (\text{equivalently } A^{-1} = A^\top).$$

Then $\|Ax\|_2 = \|x\|_2$ and $(Ax)^\top(Ay) = x^\top y$ for all x, y.

If the interviewer asks for intuition, say "orthogonal matrices preserve geometry" before you say "$A^\top A = I$". That usually lands better than giving the matrix identity with no interpretation.

Q6. Newton's method convergence speed (conceptual)

Problem. Newton iteration for $f(x) = \tanh(x^4 - x^2)$. Which root converges faster and why?

Solution. Newton's method has quadratic local convergence to a simple root where $f'(x^\star) \neq 0$. If a root is multiple or has near-zero derivative, convergence slows. Here $\tanh(u) = 0$ iff $u = 0$, so roots come from $x^4 - x^2 = x^2(x^2 - 1) = 0$:

$$x^\star \in \{-1, 0, 1\}.$$

Near $x = 0$, $u = x^4 - x^2 \approx -x^2$ is second order in x, so $f(x) \approx -x^2$ and $f'(0) = 0$, which behaves like a multiple root and leads to slower convergence. Near $x = \pm 1$, the derivative is nonzero (simple roots), so Newton converges faster there. Thus convergence is faster near $x = \pm 1$ than near $x = 0$.

Q7. $X, Y \sim \text{Unif}(0, 1)$**: compute** $\mathbb{P}(X^2 > Y)$

Problem. If X, Y are independent uniform on $(0, 1)$, find $\mathbb{P}(X^2 > Y)$.

Solution. Interpret geometrically on the unit square:

$$\mathbb{P}(X^2 > Y) = \int_0^1 \mathbb{P}(Y < x^2)\, dx = \int_0^1 x^2\, dx = \left[\frac{x^3}{3}\right]_0^1 = \frac{1}{3}.$$

12.11 Preparation Checklists

12.11.1 Case Study Checklist: Natural Gas Forecasting

Typical institution type: Commodity merchant, energy trading firm, or commodities desk.

- Understand basic natural gas market drivers (storage seasonality, weather, LNG export sensitivity).
- Be able to describe a time-series CV approach without leakage.
- Practice: build a small forecasting notebook on any commodity time series with lag features and rolling validation.
- Prepare "model upgrade" options: tree models, regime features, Bayesian shrinkage; justify only if baseline is solid.

12.11.2 General Assessment Preparation Checklist

Typical institution type: Multiple quantitative-finance institution types.

- Time pressure: practice doing clean reasoning quickly and writing concise solutions.
- Coding: Python data structures, complexity, vectorization, and debugging speed.
- Math: probability distributions, calculus (derivatives/integrals), linear algebra basics.
- Communication: in group exercises, narrate assumptions, assign roles, and keep the team aligned.

12.11.3 Trading Risk Management Preparation Checklist

Typical institution type: Investment bank, market-making firm, hedge fund, or asset manager.

- Rates basics: discount factors, bond PV, duration/convexity, swap par rate.
- Risk basics: VaR definitions + methods, stress testing, limit monitoring.
- Data: missingness handling, leakage avoidance, and time-series validation.
- Behavioral: STAR stories for teamwork conflict, motivation, and project deep dives.

12.11.4 Investment Analyst / Endowment-Style Preparation Focus

Typical institution type: Endowment, foundation, outsourced investment office, or institutional investor.

- Portfolio construction: diversification, factor exposures, drawdown and risk budgeting.
- Valuation: basic multiples, cash flow logic, understanding macro sensitivity.
- Communication: ability to summarize an investment thesis concisely and defend assumptions.

12.12 Role Maps and Preparation Priorities

Role track	Typical stages	Most tested themes	Prep priority if time is short
Commodities trading	HR screen, market case, technical round, superday, networking lunch	market intuition, time-series structure, trading motivation, fast arithmetic	build one commodity case, one Fermi framework, and one clean trading story
Quant risk analyst	fit + technical, risk modelling, stress testing, option-risk discussion	VaR, factor risk, scenario design, delta-gamma thinking	be able to give a 60-second and a 90-second VaR answer
Internal audit / tech audit	pre-recorded video interview, behaviorals, analytics tools question	controls mindset, structured communication, data tools with governance	prepare three STAR stories and one analytics-use case with safeguards
Investment management / portfolio solutions	PRVI, group preference, weakness, ethics	portfolio thinking, client constraints, judgment, remediation	have one investment or factor example and one weakness-remediation story
Quant research intern	math OA, take-home coding, pandas or modelling test	probability speed, coding completeness, debugging, notation discipline	drill timed probability sets and one pandas notebook end-to-end
Trading risk management	HR plus mixed manager rounds	swaps, bonds, rate risk, behavior under ambiguity	review rates basics and rehearse why-this-role answers with desk language
Quant technology / advisory / PM research	fixed income fundamentals, ML validation, regularization	bond math, data validation, leakage control, logistic regression intuition	build one short fixed-income explainer and one validation explainer

A useful preparation heuristic is to match every role to three buckets: what they can ask anyone, what they ask because of the desk or asset class, and what they ask because of the assessment format. Once those buckets are visible, the interview stops feeling random.

12.13 Preparation Systems

12.13.1 14-day sprint plan

Typical institution type: Multiple quantitative-finance institution types.

1. Days 1–2: inventory the target roles, map them to the companion bank, and highlight the 10 prompts or topics most likely to recur.

2. Days 3–5: build the story bank - motivation, teamwork conflict, ambiguity, weakness, ethics, and one deep technical project.
3. Days 6–8: run technical short-answer drills on VaR, swaps, bond PV, duration, validation, missing data, and logistic regression; add one 8–10 question ZQ set each day from the matching chapter.
4. Days 9–10: do one timed math set using QB/ZQ records and one timed coding or pandas set under realistic conditions.
5. Days 11–12: rehearse one market or case-study answer out loud until it becomes procedural rather than improvised.
6. Days 13–14: run two mock interviews, tighten weak openings, and prepare one-page cheat sheets for role-specific topics.

12.13.2 28-day build plan

Typical institution type: Multiple quantitative-finance institution types.

1. Week 1: foundations - use a suitable practice set for brain teasers, calculus, linear algebra, probability, and stochastic-process refresh; build the behavioral story inventory in parallel.
2. Week 2: technical frameworks - pair the QB risk/derivatives material with a suitable practice set, then add validation and desk-specific concepts.
3. Week 3: assessments - timed math, pandas, take-home simulation, and group-exercise communication drills; use a suitable practice set for algorithms and numerical methods.
4. Week 4: consolidation - mocks, role-specific reading, résumé deep dives, and day-before packs for each target firm.

12.13.3 Day-before checklist

Typical institution type: Multiple quantitative-finance institution types.

- Rehearse your opening for “tell me about yourself” and “why this role” one last time.
- Revisit only short-answer frameworks and formulas; avoid learning brand-new material.
- Prepare one example each for teamwork, conflict, fast learning, ethics, and technical depth.
- For assessment formats, decide your pacing rule in advance: when to skip, when to guess, and when to return.

12.13.4 Day-of checklist

Typical institution type: Multiple quantitative-finance institution types.

- Start every answer by naming the structure you are using: STAR, define-and-caveat, or clarify-and-decompose.
- If you are uncertain, say your assumption cleanly and continue. Ambiguity handled well scores better than silence.

- Keep a short list of anchor phrases ready: “let me define the object first”, “the main trade-off is”, “the risk is leakage”, “the desk implication is”.

12.14 Story Bank Builder

12.14.1 Core stories to prepare

Typical institution type: Cross-industry quantitative-finance employers.

Story type	What the interviewer is really probing	What your story needs
Motivation / why this role	seriousness of intent and role understanding	one real reason, one relevant example, one explicit fit statement
Conflict / slacker / disagreement	judgment and maturity under friction	diagnosis, private conversation, concrete action, outcome without drama
Weakness + remediation	self-awareness and growth rate	a real gap, a believable fix, and evidence that the fix changed your behaviour
Ethics / doing the right thing	escalation judgment and integrity	facts first, policy or mentor check, respectful escalation, documented result
Technical project deep dive	real ownership and rigor	problem, data, method, validation, impact, and one thing you would improve
Ambiguity / fast learning	comfort with incomplete information	how you framed the problem, prioritized, and learned while moving

12.14.2 One-minute answer template

Typical institution type: Cross-industry quantitative-finance employers.

1. Opening sentence: the one thing the interviewer should remember.
2. Context: only the facts needed to understand the example.
3. Your action: the decision or work you personally owned.
4. Result: quantified if possible, otherwise specific.
5. Relevance: why this example matters for the role in front of you.

12.15 Mock Interview Packs

12.15.1 Pack A: Trading / commodities

Typical institution type: Commodity merchant, energy trading firm, or commodities desk.

> Select a compact set of matching probability, trading, risk, or coding questions from your practice materials.

1. Why trading, and why this commodity in particular?
2. Walk me through the natural gas forecasting case from target definition to validation.
3. Estimate a market size or a real-world quantity with explicit assumptions.
4. A refinery explodes. What happens in the first hour, the first day, and the first week?
5. Tell me about a time you used data to make or defend a decision.

12.15.2 Pack B: Risk / fixed income

Typical institution type: Investment bank, asset manager, insurer, or institutional investor.

> Select a compact set of matching probability, trading, risk, or coding questions from your practice materials.

1. Define VaR in 60 seconds and tell me one limitation.
2. Explain how a swap is priced and what the par rate means.
3. Price a bond on the spot and tell me how coupon vs. yield affects premium or discount.
4. How would you stress test a portfolio across rates, spreads, vol, and liquidity?
5. Why this role, and what part of rate or portfolio risk interests you most?

12.15.3 Pack C: Data / ML / quant technology

Typical institution type: Cross-industry quantitative-finance employers.

> Select a compact set of matching probability, trading, risk, or coding questions from your practice materials.

1. How did you handle missing data, and why did you choose that method?
2. Explain label encoding versus one-hot encoding and when each is appropriate.
3. Walk me through your validation design and how you prevented leakage.
4. Describe one ML project end-to-end: problem, data, modelling, validation, and impact.
5. How would you explain logistic regression regularization to a non-specialist stakeholder?

12.15.4 Pack D: General Quant Foundations

Typical institution type: Multiple quantitative-finance institution types.

> Use this pack when the role is broad or the interviewer is likely to test mathematical fundamentals independent of a specific desk.

1. One structural brain teaser from a suitable practice set; explain the invariant or backward-induction step before calculating.

2. One calculus or linear-algebra question from a suitable practice set; derive rather than quote the answer.
3. Two probability questions from a suitable practice set, with at least one using conditioning or expected value.
4. One stochastic-process question from a suitable practice set; state the process, stopping rule, or martingale clearly.
5. One finance question from a suitable practice set or a role-specific derivatives or risk question.
6. One algorithm/numerical question from a suitable practice set; give correctness and complexity before implementation details.

12.16 Conclusion

Interview preparation becomes more reliable when it is treated as a system rather than a collection of memorized answers. Identify the institution type and role, classify the format, choose a compact answer structure, make assumptions explicit, and finish with an interpretation or decision. Technical depth matters, but so do pacing, error checks, and communication.

12.17 Review Questions

1. How should preparation differ among a proprietary trading firm, an investment bank, an asset manager, and a quantitative investment manager?
2. What is the shortest complete structure for a behavioral answer, and what evidence makes it credible?
3. How would you design a leakage-safe validation plan for a time-series forecasting case?
4. What should a candidate say before beginning the arithmetic in an estimation problem?
5. How do VaR, stress testing, and scenario analysis complement one another?
6. What changes when a coding task is a timed online assessment rather than a collaborative live interview?
7. Build a two-week practice schedule that combines behavioral stories, technical drills, cases, and mock interviews.

Chapter 13

Selected Interview Questions and Worked Solutions

13.1 Selection and Study Method

This chapter contains 50 representative interview questions: 35 role-linked questions and 15 cross-industry foundation questions. The selection favors recurring concepts, transferable solution methods, and coverage across trading, research, risk, investment, data, and technology roles. Repetitive variants and unusually specialized problems have been omitted.

Specific employer names and identifying details are not shown. For role-linked questions, the institution type, role, and difficulty are retained. Foundation questions are labeled for cross-industry quantitative-finance employers.

Use a three-pass method. On the first pass, answer the problem without notes and explain the setup aloud. On the second pass, compare the method with the worked solution and record the exact failure mode. On the third pass, answer a variation under time pressure. A question is not mastered merely because its solution looks familiar.

Selected area	**Questions**
Probability, puzzles, combinatorics, and decision-making	8
Statistics, machine learning, linear algebra, and time series	5
Stochastic calculus, options, and hedging	5
Risk, credit, fixed income, and portfolio management	5
Trading, market microstructure, signals, and backtesting	4
Algorithms, data structures, coding, and systems	6
Behavioral, communication, and governance	2
Cross-industry mathematical and finance foundations	15
Total	**50**

Role-Linked Questions

13.2 Probability, Puzzles, and Combinatorics

Use these questions to practice complete solutions, concise explanations, and likely follow-up questions.

13.2.1 Logic puzzles and counting arguments

Question 1 Logic / Measurement Puzzle

Institution type: Options market-making and proprietary trading firm
Role: Quantitative Trader | Difficulty: Easy

Problem. You have a 1-liter bottle and a 1.5-liter bottle. How do you measure out exactly 0.5 liters?

Solution.

Fill the 1.5 L bottle completely, then pour from it into the 1 L bottle until the 1 L bottle is full. The amount left in the 1.5 L bottle is

$$1.5 - 1.0 = 0.5 \text{ L}.$$

Core skill tested. This is a structured elimination problem rather than a formula dump. The interviewer is looking for disciplined bookkeeping: define the unknowns, translate each statement into a constraint, and use consistency to rule out impossible cases without losing track of the assumptions.

How to present it in interview. State the unknowns and the rule of the game first, then convert each sentence into a constraint you can reuse. Interviewers like hearing the consistency checks out loud because it shows that you are controlling the search space instead of guessing and backfitting.

Common traps.

- Treating each sentence in isolation instead of writing a small system of constraints.
- Forgetting that truth/lie or ordering rules depend on who is speaking to whom.
- Doing arithmetic too early before the structural consistency conditions are fixed.

Related follow-ups.

- Can you solve the same puzzle by contradiction or by ordering the cases differently?
- What is the minimum set of constraints that actually determines the unique solution?

Question 2 Puzzle / Tournament Reasoning

Institution type: Quantitative investment management firm
Role: Quantitative Researcher | Difficulty: Medium

Problem. You have 25 horses and can race 5 at a time; there is no timer (only relative ordering within a race). What is the minimum number of races needed to determine the top 3 fastest horses?

Solution.

The minimum number of races is $\boxed{7}$.

1. Split the 25 horses into 5 groups of 5, and race each group once. This takes 5 races. Let the finishing orders within the groups be

$$A_1 > A_2 > A_3 > A_4 > A_5, \quad \ldots, \quad E_1 > E_2 > E_3 > E_4 > E_5.$$

2. Race the 5 group winners A_1, B_1, C_1, D_1, E_1. Suppose the order is

$$A_1 > B_1 > C_1 > D_1 > E_1.$$

This is the 6th race.

From this, every horse in groups D and E is eliminated. Also, B_3, B_4, B_5, C_2, C_3, C_4, C_5, and A_4, A_5 cannot be in the top 3 overall.

3. The only remaining candidates for the top 3 are

$$\{A_1, A_2, A_3, B_1, B_2, C_1\}.$$

Since A_1 already beat B_1 and C_1, A_1 is definitely in the top 3. So we only need one final race among

$$\{A_2, A_3, B_1, B_2, C_1\}$$

to determine 2nd and 3rd place.

This is the 7th race.

Core skill tested. This is a structured elimination problem rather than a formula dump. The interviewer is looking for disciplined bookkeeping: define the unknowns, translate each statement into a constraint, and use consistency to rule out impossible cases without losing track of the assumptions.

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Related follow-ups.

- Can you solve the same puzzle by contradiction or by ordering the cases differently?
- What is the minimum set of constraints that actually determines the unique solution?

13.2.2 Classical probability and combinatorics

Question 3 Probability / Infinite Series

Institution type: Options market-making and proprietary trading firm
Role: Quantitative Trader | Difficulty: Easy

Problem. Two players, A and B, alternately roll a fair six-sided die. The first player to roll a 6 wins. A goes first. What is the probability that A wins?

Solution.

Let p be the probability that A wins.

A wins immediately on the first roll with probability

$$\frac{1}{6}.$$

If both A and B fail to roll a 6 in the first round, which happens with probability

$$\left(\frac{5}{6}\right)^2 = \frac{25}{36},$$

then the game resets to the original state. Therefore,

$$p = \frac{1}{6} + \frac{25}{36}p.$$

Solving,

$$p\left(1 - \frac{25}{36}\right) = \frac{1}{6} \quad \Rightarrow \quad p \cdot \frac{11}{36} = \frac{1}{6} \quad \Rightarrow \quad p = \frac{6}{11}.$$

Thus, the probability that A wins is $\boxed{\frac{6}{11}}$.

Core skill tested. This is a structured elimination problem rather than a formula dump. The interviewer is looking for disciplined bookkeeping: define the unknowns, translate each statement into a constraint, and use consistency to rule out impossible cases without losing track of the assumptions.

How to present it in interview. State the unknowns and the rule of the game first, then convert each sentence into a constraint you can reuse. Interviewers like hearing the consistency checks out loud because it shows that you are controlling the search space instead of guessing and backfitting.

Common traps.

- Treating each sentence in isolation instead of writing a small system of constraints.
- Forgetting that truth/lie or ordering rules depend on who is speaking to whom.
- Doing arithmetic too early before the structural consistency conditions are fixed.

Related follow-ups.

- Can you solve the same puzzle by contradiction or by ordering the cases differently?
- What is the minimum set of constraints that actually determines the unique solution?

Question 4 Combinatorics / Probability

Institution type: Multi-strategy alternative investment firm
Role: Quantitative Trader | Difficulty: Medium

Problem. A fair die is rolled n times. What is the probability that the sum of all rolls equals k? i.i.d.

Solution.

Let $X_1, \ldots, X_n \sim \text{Unif}\{1, 2, 3, 4, 5, 6\}$, and let

$$S_n = \sum_{i=1}^{n} X_i.$$

Clearly, $k \in \{n, \ldots, 6n\}$; otherwise, the probability is 0.

The number of integer solutions to

$$x_1 + \cdots + x_n = k, \qquad 1 \le x_i \le 6,$$

equals the coefficient of z^k in

$$(z + z^2 + \cdots + z^6)^n.$$

By inclusion–exclusion,

$$\#(k;n) = \sum_{j=0}^{\lfloor \frac{k-n}{6} \rfloor} (-1)^j \binom{n}{j} \binom{k-6j-1}{n-1}.$$

Therefore,

$$\mathbb{P}(S_n = k) = \frac{1}{6^n} \sum_{j=0}^{\lfloor \frac{k-n}{6} \rfloor} (-1)^j \binom{n}{j} \binom{k-6j-1}{n-1}, \qquad n \le k \le 6n.$$

Notes. For large n, a normal approximation works:

$$S_n \approx N\left(3.5n,\ \frac{35}{12}n\right),$$

using a continuity correction.

Core skill tested. This is a structured elimination problem rather than a formula dump. The interviewer is looking for disciplined bookkeeping: define the unknowns, translate each statement into a constraint, and use consistency to rule out impossible cases without losing track of the assumptions.

How to present it in interview. State the unknowns and the rule of the game first, then convert each sentence into a constraint you can reuse. Interviewers like hearing the consistency checks out loud because it shows that you are controlling the search space instead of guessing and backfitting.

Common traps.

- Treating each sentence in isolation instead of writing a small system of constraints.
- Forgetting that truth/lie or ordering rules depend on who is speaking to whom.
- Doing arithmetic too early before the structural consistency conditions are fixed.

Related follow-ups.

- Can you solve the same puzzle by contradiction or by ordering the cases differently?
- What is the minimum set of constraints that actually determines the unique solution?

Question 5 Randomization / Fairness

Institution type: Quantitative proprietary trading firm
Role: Quantitative Researcher | Difficulty: Easy

Problem. Design a fair lottery system using a single biased coin with unknown bias $p = P(H)$.
Solution.
Assume $0 < p < 1$. Use the von Neumann extractor.

1. Flip the biased coin twice. If the outcome is HT, output 1; if it is TH, output 0; if it is HH or TT, discard the pair and repeat.

2. Because
$$P(HT) = p(1-p) = P(TH),$$
the emitted bit is exactly fair even though p is unknown.

3. A two-flip attempt succeeds with probability $2p(1-p)$, so the expected number of original coin flips per fair bit is
$$\frac{2}{2p(1-p)} = \frac{1}{p(1-p)}.$$

To choose uniformly among N labels, generate $k = \lceil \log_2 N \rceil$ fair bits, interpret them as $U \in \{0, \dots, 2^k - 1\}$, accept if $U < N$, and otherwise reject and resample. The accepted outcome is exactly uniform on $\{0, \dots, N-1\}$.

If $p = 0$ or $p = 1$, the coin is deterministic and no procedure using only that coin can generate a non-degenerate fair random bit.

Core skill tested. This is a structured elimination problem rather than a formula dump. The interviewer is looking for disciplined bookkeeping: define the unknowns, translate each statement into a constraint, and use consistency to rule out impossible cases without losing track of the assumptions.

How to present it in interview. State the unknowns and the rule of the game first, then convert each sentence into a constraint you can reuse. Interviewers like hearing the consistency checks out loud because it shows that you are controlling the search space instead of guessing and backfitting.

Common traps.

- Treating each sentence in isolation instead of writing a small system of constraints.
- Forgetting that truth/lie or ordering rules depend on who is speaking to whom.
- Doing arithmetic too early before the structural consistency conditions are fixed.

Related follow-ups.

- Can you solve the same puzzle by contradiction or by ordering the cases differently?
- What is the minimum set of constraints that actually determines the unique solution?

Question 6 Probability

Institution type: Quantitative proprietary trading and market-making firm
Role: Quantitative Trader | Difficulty: Medium

Problem. Given a fair twelve-sided die (faces 1-12), you roll repeatedly until the cumulative sum is odd. Which odd total occurs with the highest probability?

Solution.

Let $p_s = P(S = s)$ for odd s. On the first roll, either we roll the terminal odd value immediately, or we roll an even value $e \in \{2, 4, 6, 8, 10, 12\}$ and restart from the same problem with e already added. Hence
$$p_s = \frac{1}{12}\mathbf{1}_{\{s \in \{1,3,5,7,9,11\}\}} + \frac{1}{12} \sum_{\substack{e \in \{2,4,6,8,10,12\} \\ e < s}} p_{s-e}.$$

Starting from $p_s = 0$ for $s \le 0$, this gives

$$\begin{aligned} p_1 &= 0.083333, \\ p_3 &= 0.090278, \\ p_5 &= 0.097801, \\ p_7 &= 0.105951, \\ p_9 &= 0.114780, \\ p_{11} &= 0.124345. \end{aligned}$$

These probabilities are strictly increasing through 11.

For any odd $s > 11$, the direct-stop term disappears, so

$$p_s = \frac{1}{12}\sum p_{s-e} \le \frac{6}{12}\max_{u<s} p_u.$$

Thus no value above 11 can exceed the largest probability already attained among $1, 3, 5, 7, 9, 11$. Therefore the most likely final odd total is

$$\boxed{11}.$$

Core skill tested. This is a structured elimination problem rather than a formula dump. The interviewer is looking for disciplined bookkeeping: define the unknowns, translate each statement into a constraint, and use consistency to rule out impossible cases without losing track of the assumptions.

How to present it in interview. State the unknowns and the rule of the game first, then convert each sentence into a constraint you can reuse. Interviewers like hearing the consistency checks out loud because it shows that you are controlling the search space instead of guessing and backfitting.

Common traps.

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- Forgetting that truth/lie or ordering rules depend on who is speaking to whom.
- Doing arithmetic too early before the structural consistency conditions are fixed.

Related follow-ups.

- Can you solve the same puzzle by contradiction or by ordering the cases differently?
- What is the minimum set of constraints that actually determines the unique solution?

Question 7 Probability

Institution type: Quantitative proprietary trading and market-making firm
Role: Quantitative Trader | Difficulty: Medium

Problem. A stick is broken at one random point. Then one of the pieces is chosen at random and broken at a random point. What is the probability that the three pieces can form a triangle?

Solution.

Let the first break be at

$$X \sim \mathrm{Unif}(0, 1),$$

creating pieces of lengths X and $1 - X$. Choose one of these two pieces with probability $1/2$; call its length

$$L \in \{X, 1 - X\}.$$

Now break that chosen piece at

$$U \sim \text{Unif}(0,1),$$

producing lengths

$$UL, \qquad (1-U)L, \qquad 1-L.$$

Since the total length is 1, the three pieces form a triangle if and only if no piece has length at least $1/2$, i.e.

$$\max\{UL, (1-U)L, 1-L\} < \frac{1}{2}.$$

Case 1: choose the X-piece (probability $1/2$).

We need

$$1 - X < \frac{1}{2} \qquad \Longrightarrow \qquad X > \frac{1}{2},$$

and also

$$UX < \frac{1}{2}, \qquad (1-U)X < \frac{1}{2}.$$

These give

$$U < \frac{1}{2X}, \qquad U > 1 - \frac{1}{2X}.$$

So for fixed $X > \frac{1}{2}$, the allowed interval for U is

$$1 - \frac{1}{2X} < U < \frac{1}{2X},$$

whose length is

$$\frac{1}{X} - 1.$$

Thus the contribution from this case is

$$\frac{1}{2}\int_{1/2}^{1} \left(\frac{1}{x} - 1\right) dx = \frac{1}{2}\big(\ln 2 - \tfrac{1}{2}\big).$$

Case 2: choose the $(1-X)$-piece (probability $1/2$).

By symmetry, this requires

$$X < \frac{1}{2},$$

and the allowed interval for U has length

$$\frac{1}{1-X} - 1.$$

Hence the contribution is

$$\frac{1}{2}\int_{0}^{1/2} \left(\frac{1}{1-x} - 1\right) dx = \frac{1}{2}\big(\ln 2 - \tfrac{1}{2}\big).$$

Adding the two cases,

$$\mathbb{P}(\text{triangle}) = \ln 2 - \frac{1}{2} \approx 0.1931.$$

Note. This differs from the classic problem of making two independent uniform cuts on the whole stick, which gives probability $1/4$.

Core skill tested. This is a structured elimination problem rather than a formula dump. The interviewer is looking for disciplined bookkeeping: define the unknowns, translate each statement into a constraint, and use consistency to rule out impossible cases without losing track of the assumptions.

How to present it in interview. State the unknowns and the rule of the game first, then convert each sentence into a constraint you can reuse. Interviewers like hearing the consistency checks out loud because it shows that you are controlling the search space instead of guessing and backfitting.

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Related follow-ups.

- Can you solve the same puzzle by contradiction or by ordering the cases differently?
- What is the minimum set of constraints that actually determines the unique solution?

13.2.3 Conditional probability, Bayes, and expectation

Question 8 Probability / Bayes

Institution type: Quantitative proprietary trading and market-making firm
Role: Quantitative Trader | Difficulty: Easy

Problem. You have 100 coins: 99 are fair and 1 is double-headed. You pick one uniformly at random and flip it 10 times, observing 10 heads. What is the probability that you picked the double-headed coin?

Solution.

Let D be the event "picked the double-headed coin," and let H be the event "10 heads in 10 flips."

The prior probabilities are

$$\mathbb{P}(D) = \frac{1}{100}, \qquad \mathbb{P}(D^c) = \frac{99}{100}.$$

The likelihoods are

$$\mathbb{P}(H \mid D) = 1, \qquad \mathbb{P}(H \mid D^c) = \left(\frac{1}{2}\right)^{10} = \frac{1}{1024}.$$

By Bayes' rule,

$$\mathbb{P}(D \mid H) = \frac{\mathbb{P}(H \mid D)\mathbb{P}(D)}{\mathbb{P}(H \mid D)\mathbb{P}(D) + \mathbb{P}(H \mid D^c)\mathbb{P}(D^c)}.$$

Substituting,

$$\mathbb{P}(D \mid H) = \frac{1 \cdot \frac{1}{100}}{1 \cdot \frac{1}{100} + \frac{1}{1024} \cdot \frac{99}{100}} = \frac{1}{1 + \frac{99}{1024}} = \frac{1024}{1123} \approx 0.912.$$

Therefore, the probability that you picked the double-headed coin is

$$\boxed{\frac{1024}{1123} \approx 0.912}.$$

Core skill tested. This is a pattern-recognition problem. A strong candidate identifies the smallest correct framework first, writes the governing event or quantity cleanly, and only then performs the arithmetic or algebra needed to finish the argument.

How to present it in interview. In interview conditions, announce the trick early. Say whether you are using symmetry, conditioning, a complement, an invariant, or a replication argument. That helps the interviewer follow your reasoning and reduces the chance that arithmetic clutter hides the main idea.

Common traps.

- Rushing into arithmetic before defining the relevant event or random variable.
- Missing a tie rule, support restriction, or conditioning case that changes the answer.
- Failing to sanity-check the final value against an obvious bound or symmetry argument.

Related follow-ups.

- What is the shortest alternative derivation: symmetry, complement, recursion, or indicator variables?
- How would the answer change under a biased draw, more players, or a different support?

13.3 Statistics, Machine Learning, and Econometrics

Use these questions to practice complete solutions, concise explanations, and likely follow-up questions.

13.3.1 Applied ML, feature engineering, and data quality

Question 9 Data Quality

Institution type: Global investment bank and financial services firm
Role: Risk / Quant | Difficulty: Easy

Problem. Give a step-by-step checklist for cleaning and validating financial data.
Solution.

1. **Schema and keys:** check uniqueness, referential integrity, and consistency of units and currencies.
2. **Time alignment:** verify calendars, time zones, holiday rolls, and point-in-time reference data.
3. **Missing values and outliers:** define treatment rules, apply winsorization where appropriate, and adjust for corporate actions such as splits and dividends.
4. **Reconciliation:** compare source data against the golden copy, verify roll-forwards, and reconcile totals versus subtotals.
5. **Testing and governance:** run deterministic unit tests, statistical sanity checks, and maintain an audit trail with versioning.

Core skill tested. This is a pattern-recognition problem. A strong candidate identifies the smallest correct framework first, writes the governing event or quantity cleanly, and only then performs the arithmetic or algebra needed to finish the argument.

How to present it in interview. In interview conditions, announce the trick early. Say whether you are using symmetry, conditioning, a complement, an invariant, or a replication argument. That helps the interviewer follow your reasoning and reduces the chance that arithmetic clutter hides the main idea.

Common traps.

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- Failing to sanity-check the final value against an obvious bound or symmetry argument.

Related follow-ups.

- What is the shortest alternative derivation: symmetry, complement, recursion, or indicator variables?
- How would the answer change under a biased draw, more players, or a different support?

Question 10 Data Preprocessing / Missing Data

Institution type: Quantitative investment management firm
Role: Quantitative Researcher | Difficulty: Medium

Problem. You have a dataset with missing values. How would you handle this for a machine-learning model?

Solution.

Start by diagnosing the missingness mechanism (MCAR, MAR, or plausibly MNAR) and whether missingness itself contains signal. Then build the treatment inside a leakage-safe validation pipeline.

1. **Profile missingness.** Measure rates by feature/time/entity, examine missingness indicators, and test whether missingness is associated with observed covariates or the target.

2. **Choose a method.** Simple median/mode imputation is a strong baseline; MICE, k-NN, EM, or matrix-completion methods may help when justified; many tree models can handle missing values directly.

3. **Respect time.** For forecasting, only use information that would have existed at the prediction timestamp. Forward fill can be valid with an age cap when the last observation is genuinely known; backward fill generally leaks future information and should not cross a forecasting boundary.

4. **Fit inside each fold.** Estimate imputation parameters using training data only. Never compute medians, nearest neighbors, or learned imputers using validation/test observations.

5. **Validate uncertainty and production behavior.** Compare with/without missingness indicators, monitor missing-rate drift, define all-missing fallbacks, and for multiple imputation propagate between-imputation uncertainty when inferential validity matters.

Under MNAR, no imputation algorithm automatically removes bias; sensitivity analysis or an explicit missingness model may be required.

Core skill tested. This item tests whether you can move cleanly from definition to estimation to validation. A polished response states the estimator or framework, clarifies its assumptions,

and then points out the most likely failure mode such as leakage, non-stationarity, or unstable inference.

How to present it in interview. A concise interview answer should always separate modelling from evaluation. State the model or diagnostic, then immediately describe the validation design and the reason you trust or distrust the result under realistic data conditions.

Common traps.

- Stating an estimator but not the assumptions needed for interpretation.
- Forgetting to separate train/validation/test logic or time ordering.
- Choosing a metric that does not match the business or modelling objective.

Related follow-ups.

- What breaks first if the data are non-stationary, heavy-tailed, or missing not at random?
- How would you redesign the validation scheme if the labels overlap or the data are strictly time ordered?

13.3.2 Linear models, estimation, and linear algebra

Question 11 OLS / Linear Algebra

Institution type: Cross-industry quantitative-finance employers
Role: Quant Research | Difficulty: Easy

Problem. What is the closed-form ordinary least squares (OLS) estimator for β in $\mathrm{y} = \mathrm{X}\beta + \varepsilon$?

Solution.

If

$$X \in \mathbb{R}^{n \times d}$$

has full column rank ($\operatorname{rank}(X) = d \leq n$), then the normal equations are

$$X^\top X \, \hat{\beta} = X^\top y.$$

Since $X^\top X$ is invertible in this case,

$$\hat{\beta} = (X^\top X)^{-1} X^\top y.$$

If $X^\top X$ is singular, use the Moore–Penrose pseudoinverse:

$$\hat{\beta} = X^{+} y.$$

Core skill tested. This is a structured elimination problem rather than a formula dump. The interviewer is looking for disciplined bookkeeping: define the unknowns, translate each statement into a constraint, and use consistency to rule out impossible cases without losing track of the assumptions.

How to present it in interview. State the unknowns and the rule of the game first, then convert each sentence into a constraint you can reuse. Interviewers like hearing the consistency checks out loud because it shows that you are controlling the search space instead of guessing and backfitting.

Common traps.

- Treating each sentence in isolation instead of writing a small system of constraints.

- Forgetting that truth/lie or ordering rules depend on who is speaking to whom.
- Doing arithmetic too early before the structural consistency conditions are fixed.

Related follow-ups.

- Can you solve the same puzzle by contradiction or by ordering the cases differently?
- What is the minimum set of constraints that actually determines the unique solution?

Question 12 Regression / Multicollinearity

Institution type: Cross-industry quantitative-finance employers
Role: Quant Research | Difficulty: Medium

Problem. Explain multicollinearity, how to detect it, and how ridge regression remedies it.

Solution.

Definition. Multicollinearity occurs when predictors are exactly or nearly linearly dependent. This inflates the variance of $\hat{\beta}$ and makes

$$X^\top X$$

ill-conditioned or singular.

Detection.

- inspect pairwise correlation heatmaps;
- regress each predictor X_i on the others and compute the variance inflation factor

$$\mathrm{VIF}_i = \frac{1}{1 - R_i^2};$$

  large VIF values (for example, greater than 5 or 10) indicate potential multicollinearity;
- check the condition number or the eigenvalues of $X^\top X$.

Ridge remedy. Ridge regression solves

$$\hat{\beta}_\lambda = \arg\min_\beta \ \|y - X\beta\|_2^2 + \lambda\|\beta\|_2^2.$$

The first-order condition gives

$$(X^\top X + \lambda I)\hat{\beta}_\lambda = X^\top y,$$

so

$$\hat{\beta}_\lambda = (X^\top X + \lambda I)^{-1} X^\top y.$$

Adding λI makes the system strictly positive definite for $\lambda > 0$, stabilizes inversion, and shrinks coefficient estimates toward zero, thereby reducing variance.

Core skill tested. This item tests whether you can move cleanly from definition to estimation to validation. A polished response states the estimator or framework, clarifies its assumptions, and then points out the most likely failure mode such as leakage, non-stationarity, or unstable inference.

How to present it in interview. A concise interview answer should always separate modelling from evaluation. State the model or diagnostic, then immediately describe the validation design and the reason you trust or distrust the result under realistic data conditions.

Common traps.

- Stating an estimator but not the assumptions needed for interpretation.
- Forgetting to separate train/validation/test logic or time ordering.
- Choosing a metric that does not match the business or modelling objective.

Related follow-ups.

- What breaks first if the data are non-stationary, heavy-tailed, or missing not at random?
- How would you redesign the validation scheme if the labels overlap or the data are strictly time ordered?

13.3.3 Time series, stationarity, and diagnostics

Question 13 Time Series / Unit-Root Testing
Institution type: Quantitative investment management firm
Role: Quantitative Researcher | Difficulty: Medium

Problem. You observe a time series that appears to have a unit root. How would you test for stationarity and what are the implications for modeling?

Solution.

Step 1 — Visual and preliminary checks. Inspect the ACF/PACF and rolling summary statistics. If the variance drifts or the ACF decays slowly, a unit root is plausible.

Step 2 — Augmented Dickey–Fuller (ADF) test. Estimate

$$\Delta y_t = \alpha + \beta t + \gamma y_{t-1} + \sum_{i=1}^{p} \phi_i \Delta y_{t-i} + \varepsilon_t,$$

and test

$$H_0 : \gamma = 0 \qquad \text{(unit root, } y_t \text{ is } I(1)\text{)},$$

versus

$$H_1 : \gamma < 0 \qquad \text{(stationary)}.$$

Choose the lag p by AIC or BIC so that ε_t is approximately white noise, and include the trend term βt only if the series level exhibits a deterministic trend. The ADF statistic uses non-standard Dickey–Fuller critical values.

Step 3 — Complementary checks.

- **KPSS test:** the null is (trend-)stationarity. Rejection together with ADF non-rejection strengthens evidence of a unit root.
- **Phillips–Perron (PP) test:** robust to more general serial correlation and heteroskedasticity.

Implications for modeling.

- If a unit root is confirmed, difference the series until it is stationary and model it with ARIMA(p, d, q), where d is the differencing order.

- If multiple series are $I(1)$ but cointegrated, use a vector error-correction model (VECM):

$$\Delta y_t = \Pi y_{t-1} + \sum_{i=1}^{k-1} \Gamma_i \Delta y_{t-i} + u_t, \qquad \Pi = \alpha\beta^\top,$$

  or a single-equation ECM of the form

$$\Delta y_t = \lambda(y_{t-1} - \beta^\top x_{t-1}) + \cdots .$$

- Forecasting and inference should be done on stationary transforms; level forecasts can then be re-integrated.

Core skill tested. Geometric probability questions are usually won by choosing the right picture before doing any algebra. The interviewer wants to see whether you can translate the verbal statement into a region, a symmetry argument, or a measure-preserving transform and then compute only what is necessary.

How to present it in interview. In interview conditions, announce the trick early. Say whether you are using symmetry, conditioning, a complement, an invariant, or a replication argument. That helps the interviewer follow your reasoning and reduces the chance that arithmetic clutter hides the main idea.

Common traps.

- Rushing into arithmetic before defining the relevant event or random variable.
- Missing a tie rule, support restriction, or conditioning case that changes the answer.
- Failing to sanity-check the final value against an obvious bound or symmetry argument.

Related follow-ups.

- What is the shortest alternative derivation: symmetry, complement, recursion, or indicator variables?
- How would the answer change under a biased draw, more players, or a different support?

13.4 Derivatives, Stochastic Calculus, and Option Theory

Use these questions to practice complete solutions, concise explanations, and likely follow-up questions.

13.4.1 GBM, Ito calculus, martingales, and Monte Carlo

Question 14 Stochastic Calculus

Institution type: Multi-strategy alternative investment firm
Role: Quantitative Trader / Difficulty: Medium

Problem. Derive the closed-form solution of geometric Brownian motion (GBM) and briefly explain its main applications in finance.

Solution.

Let S_t follow

$$\frac{dS_t}{S_t} = \mu\, dt + \sigma\, dW_t, \qquad S_0 > 0,$$

with constant drift μ, volatility $\sigma > 0$, and $(W_t)_{t\geq 0}$ a standard Brownian motion.

Apply Itô's lemma to $f(S_t) = \ln S_t$. Then

$$d\ln S_t = \frac{1}{S_t}\,dS_t - \frac{1}{2S_t^2}(dS_t)^2 = \left(\mu - \frac{1}{2}\sigma^2\right)dt + \sigma\,dW_t.$$

Integrating from 0 to t,

$$\ln S_t = \ln S_0 + \left(\mu - \frac{1}{2}\sigma^2\right)t + \sigma W_t.$$

Therefore,

$$S_t = S_0 \exp\left(\left(\mu - \frac{1}{2}\sigma^2\right)t + \sigma W_t\right).$$

Hence

$$\ln S_t \sim N\left(\ln S_0 + \left(\mu - \frac{1}{2}\sigma^2\right)t,\ \sigma^2 t\right),$$

so S_t is lognormal.

Its moments are

$$\mathbb{E}[S_t] = S_0 e^{\mu t},$$

and

$$\mathrm{Var}(S_t) = S_0^2 e^{2\mu t}\left(e^{\sigma^2 t} - 1\right).$$

Applications in finance.

- **Derivative pricing (Black–Scholes):** under the risk-neutral measure, replace μ by $r - q$, where r is the risk-free rate and q is the dividend yield. The GBM solution gives the lognormal terminal distribution used in Black–Scholes and Monte Carlo pricing.
- **Risk management:** simulate P&L scenarios for equities, FX, and commodities; compute risk metrics such as VaR and expected shortfall under lognormal dynamics.
- **Volatility and elasticity intuition:** log-returns are normal with variance $\sigma^2 t$, and relative changes scale with price, which avoids negative prices.
- **Portfolio problems:** continuous-time growth-optimal/Kelly analysis and Merton portfolio choice with GBM assets.

Core skill tested. The core skill here is to connect a formula or payoff identity to the economic reason it matters. Good answers do not stop at the equation; they also explain which no-arbitrage argument, SDE manipulation, or hedging interpretation makes the result useful in practice.

How to present it in interview. The clean interview version is usually a 30-to-60 second explanation: define the process or parity, write the one essential equation, and state what it implies for pricing, hedging, or risk. Avoid drowning the answer in notation unless the interviewer explicitly asks for the full derivation.

Common traps.

- Mixing up payoff, profit, and present value, or confusing physical and risk-neutral interpretations.
- Using the right formula but never saying what assumption makes it valid.
- Losing points on sign conventions, discounting, or boundary conditions.

Related follow-ups.

- How would the answer change under dividends, stochastic volatility, or a different compounding convention?
- Can you derive the same result from replication, PDE language, and intuition about convexity or hedging?

Question 15 Stochastic Calculus / Derivatives

Institution type: Multi-strategy alternative investment firm
Role: Quantitative Trader / Difficulty: Medium

Problem. What is Itô's lemma and how is it used in derivative pricing?
Solution.
Itô's lemma (one-dimensional diffusion). Let X_t satisfy

$$dX_t = a(t, X_t)\,dt + b(t, X_t)\,dW_t,$$

and let $f(t,x) \in C^{1,2}$. Then

$$df(t, X_t) = f_t\,dt + f_x\,dX_t + \frac{1}{2} f_{xx}(dX_t)^2.$$

Using

$$(dW_t)^2 = dt, \qquad dt\,dW_t = 0, \qquad dt^2 = 0,$$

this becomes

$$df(t, X_t) = \left(f_t + a f_x + \frac{1}{2} b^2 f_{xx} \right) dt + b f_x\,dW_t.$$

Use in risk-neutral derivative pricing.
Suppose the underlying follows GBM under the risk-neutral measure $\mathbb{Q}$:

$$dS_t = rS_t\,dt + \sigma S_t\,dW_t^{\mathbb{Q}}.$$

For a derivative price $V(t,S) = f(t,S)$, apply Itô's lemma:

$$dV = V_t\,dt + V_S\,dS + \frac{1}{2} V_{SS}(dS)^2.$$

Substituting dS,

$$dV = \left(V_t + rSV_S + \frac{1}{2}\sigma^2 S^2 V_{SS} \right) dt + \sigma S V_S\,dW_t^{\mathbb{Q}}.$$

Now form a self-financing delta-hedged portfolio

$$\Pi = V - \Delta S, \qquad \Delta = V_S.$$

The $dW_t^{\mathbb{Q}}$ term cancels, so Π is locally riskless and must earn the risk-free rate:

$$d\Pi = r\Pi\,dt.$$

Hence

$$V_t + rSV_S + \frac{1}{2}\sigma^2 S^2 V_{SS} - rV = 0,$$

with terminal condition

$$V(T, S) = \text{payoff}(S).$$

This is the Black–Scholes PDE. Solving it, or equivalently taking the risk-neutral expectation, gives derivative prices. For example, with payoff

$$V(T, S) = (S - K)^+,$$

the solution is the Black–Scholes call formula

$$C = SN(d_1) - Ke^{-r(T-t)}N(d_2).$$

Remarks. Itô's lemma is also used to:

- derive the lognormality of GBM via $f = \ln S$;
- obtain Greeks such as V_S and V_{SS};
- extend pricing arguments to multi-factor models via the multidimensional Itô formula with cross terms.

Core skill tested. The core skill here is to connect a formula or payoff identity to the economic reason it matters. Good answers do not stop at the equation; they also explain which no-arbitrage argument, SDE manipulation, or hedging interpretation makes the result useful in practice.

How to present it in interview. The clean interview version is usually a 30-to-60 second explanation: define the process or parity, write the one essential equation, and state what it implies for pricing, hedging, or risk. Avoid drowning the answer in notation unless the interviewer explicitly asks for the full derivation.

Common traps.

- Mixing up payoff, profit, and present value, or confusing physical and risk-neutral interpretations.
- Using the right formula but never saying what assumption makes it valid.
- Losing points on sign conventions, discounting, or boundary conditions.

Related follow-ups.

- How would the answer change under dividends, stochastic volatility, or a different compounding convention?
- Can you derive the same result from replication, PDE language, and intuition about convexity or hedging?

Question 16 Monte Carlo

Institution type: Global investment bank and financial services firm
Role: Risk / Quant | Difficulty: Medium

Problem. Describe the Monte Carlo workflow for market risk.
Solution.

1. Specify marginals and dependence, for example with a Gaussian copula or dynamic volatility model.
2. Generate M scenarios, or full paths if the problem is path-dependent.
3. Reprice positions under each scenario, using either full revaluation or a risk-factor approximation.

4. Aggregate losses and compute metrics such as VaR, expected shortfall, PFE, and CVA.

5. Improve efficiency with antithetic variates, control variates, stratified sampling, or quasi-random sequences, and perform convergence checks.

Core skill tested. This is a risk-and-finance framing problem. The best answer defines the object precisely, states the minimal formula or workflow, and then immediately mentions the practical assumption set, because interviewers care about whether you know where textbook simplifications break in a desk setting.

How to present it in interview. Answer these by sequencing definition -> construction -> caveat. For example, define the measure or sensitivity, say how you would compute it on real data or positions, and then mention one limitation such as liquidity, model error, or convention dependence.

Common traps.

- Quoting a definition without specifying horizon, confidence level, or market convention.
- Ignoring model risk, liquidity, or scenario design when moving from textbook to practice.
- Giving a sensitivity or valuation formula without saying what inputs must be bootstrapped or estimated.

Related follow-ups.

- What part of the workflow is convention-dependent, and what part is model-dependent?
- How would you explain the same concept to a portfolio manager instead of a quant interviewer?

13.4.2 Options, no-arbitrage, Black-Scholes, Greeks, and hedging

Question 17 Black–Scholes

Institution type: Global investment bank and financial services firm
Role: Risk / Quant | Difficulty: Easy

Problem. Summarize the Black-Scholes framework, key assumptions, and pricing steps.
Solution.
Under Black–Scholes–Merton with continuous dividend yield q,

$$\frac{dS_t}{S_t} = (r - q)\,dt + \sigma\,dW_t^{\mathbb{Q}}$$

under the risk-neutral measure. The model assumes no arbitrage, frictionless continuous trading in the idealization, constant r, q, σ, and lognormal diffusion dynamics.

For a European call,

$$C = S_0 e^{-qT}\Phi(d_1) - Ke^{-rT}\Phi(d_2),$$

where

$$d_1 = \frac{\ln(S_0/K) + (r - q + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \qquad d_2 = d_1 - \sigma\sqrt{T}.$$

The put price follows from

$$P = Ke^{-rT}\Phi(-d_2) - S_0 e^{-qT}\Phi(-d_1)$$

or from put–call parity

$$C - P = S_0 e^{-qT} - K e^{-rT}.$$

For a non-dividend-paying stock set $q = 0$.

Implementation steps: validate inputs and units, compute d_1, d_2, evaluate the normal CDF, price the option, and sanity-check intrinsic-value bounds, limiting cases, and put–call parity. The major practical caveat is that market implied volatility varies by strike and maturity, so constant σ is an idealization rather than an empirical surface model.

Core skill tested. The core skill here is to connect a formula or payoff identity to the economic reason it matters. Good answers do not stop at the equation; they also explain which no-arbitrage argument, SDE manipulation, or hedging interpretation makes the result useful in practice.

How to present it in interview. The clean interview version is usually a 30-to-60 second explanation: define the process or parity, write the one essential equation, and state what it implies for pricing, hedging, or risk. Avoid drowning the answer in notation unless the interviewer explicitly asks for the full derivation.

Common traps.

- Mixing up payoff, profit, and present value, or confusing physical and risk-neutral interpretations.
- Using the right formula but never saying what assumption makes it valid.
- Losing points on sign conventions, discounting, or boundary conditions.

Related follow-ups.

- How would the answer change under dividends, stochastic volatility, or a different compounding convention?
- Can you derive the same result from replication, PDE language, and intuition about convexity or hedging?

Question 18 Greeks

Institution type: Global investment bank and financial services firm
Role: Risk / Quant | Difficulty: Easy

Problem. Define the main Greeks and how you would use them in hedging.

Solution.

Let τ denote *time to expiry*. For a Black–Scholes–Merton call with dividend yield q, the main Greeks include

$$\Delta = \frac{\partial C}{\partial S} = e^{-q\tau}\Phi(d_1),$$

$$\Gamma = \frac{\partial^2 C}{\partial S^2} = \frac{e^{-q\tau}\phi(d_1)}{S\sigma\sqrt{\tau}},$$

$$\text{Vega} = \frac{\partial C}{\partial \sigma} = S e^{-q\tau}\phi(d_1)\sqrt{\tau},$$

and

$$\rho = \frac{\partial C}{\partial r} = K\tau e^{-r\tau}\Phi(d_2).$$

Be explicit about the theta convention. Market *theta* is usually the derivative with respect to calendar time t while expiry date is fixed:

$$\Theta = \frac{\partial C}{\partial t} = -\frac{\partial C}{\partial \tau}.$$

So calling $\partial C/\partial \tau$ "time decay" without the minus sign reverses the usual convention.

In hedging, use the underlying to neutralize delta, options to control gamma and vega, and rate instruments when rho matters. Rebalance according to time/trigger rules and measure residual risk, transaction costs, discrete-hedging error, skew/term-structure exposure, and model risk.

Core skill tested. The core skill here is to connect a formula or payoff identity to the economic reason it matters. Good answers do not stop at the equation; they also explain which no-arbitrage argument, SDE manipulation, or hedging interpretation makes the result useful in practice.

How to present it in interview. The clean interview version is usually a 30-to-60 second explanation: define the process or parity, write the one essential equation, and state what it implies for pricing, hedging, or risk. Avoid drowning the answer in notation unless the interviewer explicitly asks for the full derivation.

Common traps.

- Mixing up payoff, profit, and present value, or confusing physical and risk-neutral interpretations.
- Using the right formula but never saying what assumption makes it valid.
- Losing points on sign conventions, discounting, or boundary conditions.

Related follow-ups.

- How would the answer change under dividends, stochastic volatility, or a different compounding convention?
- Can you derive the same result from replication, PDE language, and intuition about convexity or hedging?

13.5 Risk, Fixed Income, and Portfolio Management

Use these questions to practice complete solutions, concise explanations, and likely follow-up questions.

13.5.1 Risk measures, stress testing, and volatility

Question 19 Value at Risk

Institution type: Global investment bank and financial services firm
Role: Risk / Quant | Difficulty: Easy

Problem. Explain Value at Risk (VaR) and its limitations. Provide a step-by-step way to compute it.

Solution.

1. **Definition.** If L denotes portfolio loss, VaR at confidence level α is the α-quantile of the loss distribution:
$$\mathrm{VaR}_\alpha = \inf\{\ell : P(L \le \ell) \ge \alpha\}.$$
For a continuous distribution this is equivalent to $P(L > \mathrm{VaR}_\alpha) = 1 - \alpha$.

2. **Historical simulation.** Revalue the current portfolio under historical shocks (or use horizon-matched historical P&L), convert results to losses, sort losses from smallest to largest, and take the empirical α-quantile. Thus 99% VaR is the 99th percentile of loss, not the 1st percentile.

3. **Delta-normal VaR.** If
$$\Delta P \approx \Delta^\top \Delta X, \qquad \Delta X \sim N(0, \Sigma),$$
then the loss standard deviation is $\sqrt{\Delta^\top \Sigma \Delta}$, and for zero mean
$$\mathrm{VaR}_\alpha = \Phi^{-1}(\alpha)\sqrt{\Delta^\top \Sigma \Delta}.$$
Add the mean term if it is material and use delta-gamma or full revaluation for nonlinear portfolios.

4. **Monte Carlo VaR.** Simulate joint risk-factor scenarios, reprice the portfolio, convert P&L to loss, and again take the empirical α-quantile of loss.

Limitations include model and parameter risk, liquidity assumptions, procyclicality, and the fact that VaR says nothing about the average severity beyond the quantile. Expected Shortfall addresses the last issue and is coherent under standard conditions, whereas VaR need not be subadditive.

Core skill tested. This is a risk-and-finance framing problem. The best answer defines the object precisely, states the minimal formula or workflow, and then immediately mentions the practical assumption set, because interviewers care about whether you know where textbook simplifications break in a desk setting.

How to present it in interview. Answer these by sequencing definition -> construction -> caveat. For example, define the measure or sensitivity, say how you would compute it on real data or positions, and then mention one limitation such as liquidity, model error, or convention dependence.

Common traps.

- Quoting a definition without specifying horizon, confidence level, or market convention.
- Ignoring model risk, liquidity, or scenario design when moving from textbook to practice.
- Giving a sensitivity or valuation formula without saying what inputs must be bootstrapped or estimated.

Related follow-ups.

- What part of the workflow is convention-dependent, and what part is model-dependent?
- How would you explain the same concept to a portfolio manager instead of a quant interviewer?

Question 20 Stress Testing

Institution type: Global investment bank and financial services firm
Role: Risk / Quant | Difficulty: Medium

Problem. How do you design stress scenarios?
Solution.

1. **Scenario type.** Use historical replay (for example, the GFC), hypothetical shocks, and reverse stress tests to identify breaking points.

2. **Scenario construction.** Apply joint shocks that respect cross-asset relationships, basis effects, and liquidity horizons.

3. **Calibration.** Anchor severity to the firm's risk appetite, historical precedent, and regulatory requirements.

4. **Outputs.** Report P&L by factor and desk, breached limits, management actions, and governance sign-off.

Core skill tested. This is a risk-and-finance framing problem. The best answer defines the object precisely, states the minimal formula or workflow, and then immediately mentions the practical assumption set, because interviewers care about whether you know where textbook simplifications break in a desk setting.

How to present it in interview. Answer these by sequencing definition -> construction -> caveat. For example, define the measure or sensitivity, say how you would compute it on real data or positions, and then mention one limitation such as liquidity, model error, or convention dependence.

Common traps.

- Quoting a definition without specifying horizon, confidence level, or market convention.
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Related follow-ups.

- What part of the workflow is convention-dependent, and what part is model-dependent?
- How would you explain the same concept to a portfolio manager instead of a quant interviewer?

13.5.2 Credit and counterparty risk

Question 21 Credit Risk Modelling

Institution type: Global investment bank and financial services firm
Role: Risk / Quant | Difficulty: Medium

Problem. Outline a step-by-step build of a loan portfolio credit risk model.
Solution.

1. **Data.** Gather borrower financials, collateral information, vintage and performance data, and macroeconomic variables. Define default and cure consistently.

2. **PD model.** Build a logistic or survival (hazard) model with borrower and macro covariates; calibrate TTC versus PIT; validate with AUC, calibration, and stability tests.

3. **LGD model.** Use beta regression, GLMs, or quantile regression based on collateral, seniority, and LTV; constrain predictions to $(0, 1)$; validate with pinball loss and calibration diagnostics.

4. **EAD / CCF.** For revolvers, model drawdown at default; for amortizing loans, project the exposure path.

5. **Portfolio aggregation.** Use factor or copula dependence structures to simulate joint defaults and compute expected loss, unexpected loss, full loss distributions, and concentration metrics.

6. **Stress testing.** Apply macro scenarios and report impacts on capital, provisions, and loss distributions.

Core skill tested. This is a risk-and-finance framing problem. The best answer defines the object precisely, states the minimal formula or workflow, and then immediately mentions the practical assumption set, because interviewers care about whether you know where textbook simplifications break in a desk setting.

How to present it in interview. Answer these by sequencing definition -> construction -> caveat. For example, define the measure or sensitivity, say how you would compute it on real data or positions, and then mention one limitation such as liquidity, model error, or convention dependence.

Common traps.

- Quoting a definition without specifying horizon, confidence level, or market convention.
- Ignoring model risk, liquidity, or scenario design when moving from textbook to practice.
- Giving a sensitivity or valuation formula without saying what inputs must be bootstrapped or estimated.

Related follow-ups.

- What part of the workflow is convention-dependent, and what part is model-dependent?
- How would you explain the same concept to a portfolio manager instead of a quant interviewer?

13.5.3 Fixed income, duration, and rates concepts

Question 22 Duration

Institution type: Global investment bank and financial services firm
Role: Risk / Quant | Difficulty: Easy

Problem. Define duration variants and how to use them.

Solution.

Macaulay duration is the present-value-weighted average time to receive the bond's cash flows.

Modified duration is the first-order sensitivity of bond price to a parallel yield shift. Under m-times compounding,

$$D_{\text{mod}} = \frac{D_{\text{Mac}}}{1 + y/m}.$$

DV01 is approximately

$$\text{DV01} \approx P \cdot D_{\text{mod}} \cdot 0.0001.$$

Key-rate durations isolate sensitivity to selected maturities on the yield curve and are useful for curve hedging.

In practice, duration is used for immunization, asset-liability matching, hedge design, and convexity-aware interest-rate risk management.

Core skill tested. This is a risk-and-finance framing problem. The best answer defines the object precisely, states the minimal formula or workflow, and then immediately mentions the practical assumption set, because interviewers care about whether you know where textbook simplifications break in a desk setting.

How to present it in interview. Answer these by sequencing definition -> construction -> caveat. For example, define the measure or sensitivity, say how you would compute it on real data or positions, and then mention one limitation such as liquidity, model error, or convention dependence.

Common traps.

- Quoting a definition without specifying horizon, confidence level, or market convention.
- Ignoring model risk, liquidity, or scenario design when moving from textbook to practice.
- Giving a sensitivity or valuation formula without saying what inputs must be bootstrapped or estimated.

Related follow-ups.

- What part of the workflow is convention-dependent, and what part is model-dependent?
- How would you explain the same concept to a portfolio manager instead of a quant interviewer?

13.5.4 Portfolio construction, factor models, and attribution

Question 23 Portfolio Optimization

Institution type: Quantitative investment management firm
Role: Quantitative Researcher / Difficulty: Medium

Problem. You have n assets with expected returns $\mu \in \mathbb{R}^n$ and covariance $\Sigma \succ 0$. Find the minimum-variance portfolio for a target expected return r (and full investment).

Solution.

Solve

$$\min_w \; w^\top \Sigma w$$

subject to

$$\mu^\top w = r, \qquad \mathbf{1}^\top w = 1.$$

The Lagrangian is

$$L(w, \lambda, \nu) = w^\top \Sigma w - \lambda(\mu^\top w - r) - \nu(\mathbf{1}^\top w - 1).$$

The first-order condition is

$$2\Sigma w - \lambda\mu - \nu\mathbf{1} = 0,$$

so

$$w = \frac{1}{2}\Sigma^{-1}(\lambda\mu + \nu\mathbf{1}).$$

Define

$$A = \mathbf{1}^\top\Sigma^{-1}\mathbf{1}, \qquad B = \mathbf{1}^\top\Sigma^{-1}\mu, \qquad C = \mu^\top\Sigma^{-1}\mu, \qquad \Delta = AC - B^2.$$

Imposing the two constraints gives

$$\lambda = \frac{2(Ar - B)}{\Delta}, \qquad \nu = \frac{2(C - Br)}{\Delta}.$$

Hence the optimal weights are

$$w^* = \Sigma^{-1}\left(\frac{Ar - B}{\Delta}\mu + \frac{C - Br}{\Delta}\mathbf{1}\right).$$

The minimum variance achieved at target return r is

$$\sigma^2(r) = \frac{Ar^2 - 2Br + C}{\Delta}.$$

Core skill tested. This is a risk-and-finance framing problem. The best answer defines the object precisely, states the minimal formula or workflow, and then immediately mentions the practical assumption set, because interviewers care about whether you know where textbook simplifications break in a desk setting.

How to present it in interview. Answer these by sequencing definition -> construction -> caveat. For example, define the measure or sensitivity, say how you would compute it on real data or positions, and then mention one limitation such as liquidity, model error, or convention dependence.

Common traps.

- Quoting a definition without specifying horizon, confidence level, or market convention.
- Ignoring model risk, liquidity, or scenario design when moving from textbook to practice.
- Giving a sensitivity or valuation formula without saying what inputs must be bootstrapped or estimated.

Related follow-ups.

- What part of the workflow is convention-dependent, and what part is model-dependent?
- How would you explain the same concept to a portfolio manager instead of a quant interviewer?

13.6 Trading, Market Making, and Statistical Arbitrage

Use these questions to practice complete solutions, concise explanations, and likely follow-up questions.

13.6.1 Market making, hedging, and microstructure

Question 24 Market Microstructure
Institution type: Quantitative proprietary trading firm
Role: Quantitative Researcher | Difficulty: Medium

Problem. Explain the concept of market microstructure and its impact on trading.
Solution.

- **Core elements:** limit-order book, tick size, queue priority (price-time), maker-taker fees, hidden or iceberg orders, and latency.
- **Empirical effects:** bid-ask bounce (which induces negative first-order autocorrelation in transaction prices), price impact Λ, the spread-volatility relation, and variance-ratio distortions at high frequency.
- **Adverse selection and impact models:**
  - Kyle model:
$$\Delta P = \Lambda Q + \eta;$$
  - Almgren–Chriss: temporary and permanent impact with optimal execution;
  - Hasbrouck-style trade-sign models for impact decay.

- **Trading implications:** choose the execution style (limit order versus market order), manage participation rate and queue position, avoid toxic flow, and measure implementation shortfall relative to benchmarks such as open, close, or VWAP.
- **Backtesting caution:** use midquotes or quote-consistent fills, and apply realistic queue, latency, and partial-fill rules. Microstructure determines costs, slippage, and execution risk, so modeling it accurately is essential for alpha preservation and capacity estimation.

Core skill tested. This prompt is less about memorizing a single formula and more about demonstrating trading judgment. You want to show that signal generation, execution costs, inventory control, and validation all live in the same answer rather than appearing as disconnected bullet points.

How to present it in interview. Treat this like a small investment memo: define the edge, state how you would implement it, then explain how you would keep yourself from being fooled by costs, stale signals, or biased backtests. The interviewer is usually looking for disciplined judgment rather than a single clever sentence.

Common traps.

- Talking about signal quality without discussing transaction costs, latency, or inventory risk.
- Describing a backtest that implicitly leaks future information or overfits thresholds.
- Presenting a strategy idea without a falsification plan or risk control.

Related follow-ups.

- What is the falsification test that would make you stop trusting the signal or quoting rule?
- How would you adapt the idea if spreads widen, liquidity disappears, or the regime changes?

Question 25 Market Making / Bayesian Pricing

Institution type: Quantitative proprietary trading and market-making firm
Role: Quantitative Trader / Difficulty: Medium

Problem. Market making scenario: A stock's "true value" V is modeled as $V \sim N(\mu, \sigma^2)$ with $\mu = 100$, $\sigma = 2$. You must quote bid/ask prices. What would you quote?

Solution.

The prior $V \sim N(100, 2^2)$ determines a natural *center* for the quotes, but it does not determine a unique bid–ask spread.

If the market maker is risk-neutral, has no inventory, faces no fees or latency, and incoming orders are pure noise independent of V, competition drives the zero-profit quotes to

$$\text{bid} = \text{ask} = \mathbb{E}[V] = 100.$$

A positive spread requires an additional economic friction or model: adverse selection, inventory risk, order-processing cost, tick size, latency risk, or a fill-intensity trade-off.

For example, in an adverse-selection model the competitive zero-profit quotes take the form

$$\text{ask} = \mathbb{E}[V \mid \text{incoming buy}], \qquad \text{bid} = \mathbb{E}[V \mid \text{incoming sell}],$$

and these conditional expectations cannot be computed until $P(\text{order side} \mid V)$ is specified.

Therefore the interview-quality answer is:

- anchor the mid at 100 under the symmetric prior;

- state that the spread is *not identified* from μ and σ alone;
- ask for, or explicitly assume, a fill/adverse-selection/cost model before quoting a numerical bid and ask.

A confidence interval such as $100 \pm z\sigma$ may be useful as a risk diagnostic, but it is not by itself a zero-profit market-making quote.

Core skill tested. This prompt is less about memorizing a single formula and more about demonstrating trading judgment. You want to show that signal generation, execution costs, inventory control, and validation all live in the same answer rather than appearing as disconnected bullet points.

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- Describing a backtest that implicitly leaks future information or overfits thresholds.
- Presenting a strategy idea without a falsification plan or risk control.

Related follow-ups.

- What is the falsification test that would make you stop trusting the signal or quoting rule?
- How would you adapt the idea if spreads widen, liquidity disappears, or the regime changes?

13.6.2 Statistical arbitrage, signals, and backtesting

Question 26 Backtesting / Experimental Design

Institution type: Quantitative proprietary trading firm
Role: Quantitative Researcher | Difficulty: Medium

Problem. How do you properly backtest a trading strategy?
Solution.

1. **Data hygiene.** Use point-in-time data; handle survivorship bias, delistings, stale quotes, corporate actions, and dividends. Eliminate look-ahead bias and leakage.
2. **Costs and execution.** Include realistic transaction costs, commissions, fees, slippage, borrow and financing costs, shorting constraints, and an execution model with market impact.
3. **Sampling and latency.** Respect exchange timestamps, signal-construction delays, order-placement latency, and executable prices rather than idealized prints.
4. **Evaluation design.** Use train/validation/test splits or walk-forward evaluation. Freeze hyperparameters before the test period and report out-of-sample performance only.
5. **Robustness.** Perform parameter sweeps, perturb costs and latency assumptions, break results down by subperiod and regime, and use bootstrap confidence intervals or reality checks.

6. **Multiple testing control.** Control data snooping using White's Reality Check, SPA tests, FDR control, or clean holdout procedures.

7. **Risk and capacity.** Monitor drawdowns, turnover, exposure limits, liquidity filters such as ADV or participation caps, concentration, and borrow availability.

8. **Reporting.** Report metrics such as IC, IR, Sharpe, Sortino, maximum drawdown, hit rate, skew, kurtosis, and t-statistics with Newey–West corrections for autocorrelation and heteroskedasticity.

Core skill tested. This prompt is less about memorizing a single formula and more about demonstrating trading judgment. You want to show that signal generation, execution costs, inventory control, and validation all live in the same answer rather than appearing as disconnected bullet points.

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Common traps.

- Talking about signal quality without discussing transaction costs, latency, or inventory risk.
- Describing a backtest that implicitly leaks future information or overfits thresholds.
- Presenting a strategy idea without a falsification plan or risk control.

Related follow-ups.

- What is the falsification test that would make you stop trusting the signal or quoting rule?
- How would you adapt the idea if spreads widen, liquidity disappears, or the regime changes?

Question 27 Backtesting / Evaluation Metrics

Institution type: Quantitative investment management firm
Role: Quantitative Researcher | Difficulty: Medium

Problem. Describe how you would backtest a trading strategy and which metrics you would use to evaluate performance.

Solution.

Protocol.

1. **Data hygiene.** Use survivorship-bias-free universes, point-in-time fundamentals, correct timestamp alignment, corporate-action adjustments, and realistic borrow or financing availability.

2. **Split design.** Use expanding or rolling walk-forward splits:

$$\text{train} \rightarrow \text{validate} \rightarrow \text{test}.$$

Fit all feature engineering and preprocessing only on in-sample data to avoid leakage.

3. **Execution model.** Include realistic commissions, fees, spreads, slippage, latency, market impact, borrow fees, funding costs, and shorting constraints.

4. **Portfolio rules.** Define sizing, leverage, turnover constraints, concentration limits, exposure caps, and rebalancing schedules.

5. **Risk control.** Add pre-trade limits, post-trade kill switches, scenario tests, and regime segmentation.

Key metrics.

- **Performance:** CAGR, Sharpe, Sortino, information ratio, CAPM alpha and beta, hit rate, average win/loss, and profit factor.
- **Risk:** volatility, maximum drawdown, Calmar ratio, VaR, Expected Shortfall, and exposures by sector, country, or factor.
- **Stability and robustness:** rolling Sharpe, turnover, capacity versus impact, parameter sensitivity, and P&L by time, volatility, or liquidity bucket.
- **Multiple testing control:** FDR or FWER control across many ideas, together with live-versus-backtest decay tracking.

Core skill tested. This prompt is less about memorizing a single formula and more about demonstrating trading judgment. You want to show that signal generation, execution costs, inventory control, and validation all live in the same answer rather than appearing as disconnected bullet points.

How to present it in interview. Treat this like a small investment memo: define the edge, state how you would implement it, then explain how you would keep yourself from being fooled by costs, stale signals, or biased backtests. The interviewer is usually looking for disciplined judgment rather than a single clever sentence.

Common traps.

- Talking about signal quality without discussing transaction costs, latency, or inventory risk.
- Describing a backtest that implicitly leaks future information or overfits thresholds.
- Presenting a strategy idea without a falsification plan or risk control.

Related follow-ups.

- What is the falsification test that would make you stop trusting the signal or quoting rule?
- How would you adapt the idea if spreads widen, liquidity disappears, or the regime changes?

13.7 Algorithms, Data Structures, and Interview Coding

Use these questions to practice complete solutions, concise explanations, and likely follow-up questions.

13.7.1 Algorithms, data structures, and coding patterns

Question 28 Data Structures / Streaming Median

Institution type: Options market-making and proprietary trading firm
Role: Quantitative Trader | Difficulty: Medium

Problem. Given a stream of numbers, design a data structure that can return the median after each insertion, efficiently.

Solution.

Maintain two heaps:

- a max-heap L for the lower half of the data,
- a min-heap R for the upper half.

Keep the invariants

$$|\text{len}(L) - \text{len}(R)| \le 1, \qquad \max(L) \le \min(R).$$

On insertion:

1. insert the new value into the appropriate heap;
2. rebalance if one heap exceeds the other by 2 elements.

The median is then:

$$\begin{aligned} &\max(L) \quad \text{if } L \text{ has one extra element,} \\ &\min(R) \quad \text{if } R \text{ has one extra element,} \end{aligned}$$

and

$$\frac{\max(L) + \min(R)}{2} \quad \text{if the heaps have equal size.}$$

Complexities:

$$O(\log n) \quad \text{per insertion,} \qquad O(1) \quad \text{per median query,} \qquad O(n) \quad \text{space.}$$

Core skill tested. This is really an invariants-and-complexity question disguised as implementation. The interviewer wants to hear what state you maintain, why that state is sufficient, and how each operation preserves the invariant while meeting the target time complexity.

How to present it in interview. Present these problems in three moves: name the invariant, describe insert/update/query, and give time and space complexity. If there is a balancing rule or a lazy-deletion rule, say it before discussing edge cases so the interviewer can see the design is principled rather than improvised.

Common traps.

- Explaining the container choice but not the invariant that makes the query correct.
- Forgetting duplicates, empty-state behaviour, or rebalance conditions.
- Omitting time and space complexity or misstating amortized versus worst-case cost.

Related follow-ups.

- Can you support deletion, duplicates, or sliding-window updates without breaking the complexity target?
- How would the design change if the stream were distributed across shards or needed persistence?

Question 29 Algorithms / DP

Institution type: Quantitative proprietary trading firm
Role: Quantitative Researcher | Difficulty: Medium

Problem. Given an integer array, find the maximum sum over all contiguous subarrays (Kadane's algorithm). Now modify it for the case where you may remove at most one element from the chosen subarray. Return the maximum attainable sum.

Solution.

Maintain two DP states at position i, both for subarrays ending at i:

$$\text{noDel}[i] := \text{maximum subarray sum ending at } i \text{ with no deletion},$$

$$\text{oneDel}[i] := \text{maximum subarray sum ending at } i \text{ with at most one deletion}.$$

For $i \geq 1$, letting $x = a_i$, the transitions are

$$\text{noDel}[i] = \max\{x,\ \text{noDel}[i-1] + x\},$$

$$\text{oneDel}[i] = \max\{\text{noDel}[i-1],\ \text{oneDel}[i-1] + x\}.$$

Interpretation:

- For noDel, either start a new subarray at x or extend the previous one.
- For oneDel, either delete the current element x and keep noDel$[i-1]$, or keep x and extend a subarray that has already used one deletion.

Initialization:

$$\text{noDel}[0] = a_0, \qquad \text{oneDel}[0] = -\infty.$$

The answer is

$$\max_i \max\{\text{noDel}[i],\ \text{oneDel}[i]\}.$$

This runs in

$$O(n)$$

time and

$$O(1)$$

space if only rolling values are stored.

Reference implementation (Python, $O(n)$ time, $O(1)$ space):

```
def max_subarray_with_one_deletion(a):
    no_del = a[0]
    one_del = float('-inf')
    ans = a[0]

    for x in a[1:]:
        one_del = max(no_del, one_del + x)   # delete x OR keep x after prior deletion
        no_del = max(x, no_del + x)          # classic Kadane step
        ans = max(ans, no_del, one_del)

    return ans
```

Core skill tested. This is really an invariants-and-complexity question disguised as implementation. The interviewer wants to hear what state you maintain, why that state is sufficient, and how each operation preserves the invariant while meeting the target time complexity.

How to present it in interview. Present these problems in three moves: name the invariant, describe insert/update/query, and give time and space complexity. If there is a balancing rule or a lazy-deletion rule, say it before discussing edge cases so the interviewer can see the design is principled rather than improvised.

Common traps.

- Explaining the container choice but not the invariant that makes the query correct.
- Forgetting duplicates, empty-state behaviour, or rebalance conditions.
- Omitting time and space complexity or misstating amortized versus worst-case cost.

Related follow-ups.

- Can you support deletion, duplicates, or sliding-window updates without breaking the complexity target?
- How would the design change if the stream were distributed across shards or needed persistence?

Question 30 Algorithms / Hashing

Institution type: Cross-industry quantitative-finance employers
Role: Quant Research | Difficulty: Medium

Problem. Given a list of words, group anagrams together (one line per group). Design and justify an $O(nL)$ method where L is average word length.

Solution.

Two strings are anagrams if and only if they share the same canonical signature.

There are two common choices:

- **Sorted signature:** sort the characters of each word. This costs

$$O(L \log L)$$

per word.

- **Count signature (preferred):** count characters into a fixed-length vector (for example, 26 entries for lowercase a-z) and use the resulting count tuple as the hash key. This costs

$$O(L)$$

per word.

Using the count signature, build a hash map

$$\text{signature} \mapsto \text{list of words},$$

which gives total complexity

$$O(nL).$$

Reference implementation (Python, count-key, case-insensitive):

```
from collections import defaultdict

def group_anagrams(words):
    buckets = defaultdict(list)
    for w in words:
        cnt = [0]*26
        for ch in w.lower():
            if 'a' <= ch <= 'z':
                cnt[ord(ch)-97] += 1
        buckets[tuple(cnt)].append(w)
    return list(buckets.values())

# Example:
# groups = group_anagrams(["earth","heart","ab","cat","listen","silent"])
# for g in groups: print(" ".join(g))
```

Core skill tested. This is really an invariants-and-complexity question disguised as implementation. The interviewer wants to hear what state you maintain, why that state is sufficient, and how each operation preserves the invariant while meeting the target time complexity.

How to present it in interview. Present these problems in three moves: name the invariant, describe insert/update/query, and give time and space complexity. If there is a balancing rule or a lazy-deletion rule, say it before discussing edge cases so the interviewer can see the design is principled rather than improvised.

Common traps.

- Explaining the container choice but not the invariant that makes the query correct.
- Forgetting duplicates, empty-state behaviour, or rebalance conditions.
- Omitting time and space complexity or misstating amortized versus worst-case cost.

Related follow-ups.

- Can you support deletion, duplicates, or sliding-window updates without breaking the complexity target?
- How would the design change if the stream were distributed across shards or needed persistence?

Question 31 Algorithms / Sliding Window

Institution type: Quantitative investment management firm
Role: Quantitative Researcher | Difficulty: Easy

Problem. Implement a function to compute the moving average of a stream of integers given a window size k.

Solution.

Maintain:

- a queue of at most k items,
- a running sum of the values currently in the queue.

When a new value arrives:

1. push the new value into the queue;

2. add it to the running sum;
3. if the queue length exceeds k, pop the oldest value and subtract it from the sum.

The current moving average is then

$$\frac{\text{sum}}{\text{len(queue)}}.$$

Reference implementation:

```
from collections import deque

class MovingAverage:
    def __init__(self, k: int):
        self.k = k
        self.q = deque()
        self.s = 0.0

    def add(self, x: float) -> float:
        self.q.append(x)
        self.s += x
        if len(self.q) > self.k:
            self.s -= self.q.popleft()
        return self.s / len(self.q)
```

Time per update:

$$O(1),$$

space:

$$O(k).$$

Core skill tested. This is really an invariants-and-complexity question disguised as implementation. The interviewer wants to hear what state you maintain, why that state is sufficient, and how each operation preserves the invariant while meeting the target time complexity.

How to present it in interview. Present these problems in three moves: name the invariant, describe insert/update/query, and give time and space complexity. If there is a balancing rule or a lazy-deletion rule, say it before discussing edge cases so the interviewer can see the design is principled rather than improvised.

Common traps.

- Explaining the container choice but not the invariant that makes the query correct.
- Forgetting duplicates, empty-state behaviour, or rebalance conditions.
- Omitting time and space complexity or misstating amortized versus worst-case cost.

Related follow-ups.

- Can you support deletion, duplicates, or sliding-window updates without breaking the complexity target?
- How would the design change if the stream were distributed across shards or needed persistence?

13.8 Systems, Order Books, and Production Prototypes

Use these questions to practice complete solutions, concise explanations, and likely follow-up questions.

13.8.1 Real-time risk, OMS, and system architecture

Question 32 Systems Design / Real-Time Risk

Institution type: Electronic market-making firm
Role: Quantitative Analyst / Difficulty: Hard

Problem. Design a real-time risk monitoring system that tracks positions across 200+ instruments and triggers alerts when limits are breached. The system must handle ~ 106 position updates per second.

Solution.

Ingestion.

- Multiple feed handlers (per venue/desk) parse updates

$$\langle ts,\ inst_id,\ acct,\ \Delta q,\ p,\ \gamma\text{-vector}\rangle.$$

- Each handler writes to an SPSC ring buffer (lock-free) consumed by the Risk Engine.
- Sequence numbers provide idempotency, and duplicates are dropped.

State (per instrument, optionally per account).

$$q_i \quad \text{current position (shares)},$$

$$n_i = p_i q_i \quad \text{notional},$$

$$\text{Greeks}_i \approx J_i q_i \quad \text{linearized risk (e.g. DV01, Vega)}.$$

All arrays are preallocated and keyed by dense

$$inst_id \in \{0, \ldots, M-1\}$$

to allow $O(1)$ updates.

Limits store. Thresholds per instrument/account:

$$|q_i| \le Q_i^{\max}, \qquad |n_i| \le N_i^{\max}, \qquad |DV01_i| \le D_i^{\max}, \qquad |Vega_i| \le V_i^{\max}.$$

Use hysteresis ($\pm\varepsilon$) and debounce windows to avoid alert chatter.

Risk Engine.

- Reads batches from the rings, applies Δq to arrays, recomputes cheap linear metrics, and compares them to limits.
- Breaches are pushed to an Alert Queue (SPSC), consumed by an out-of-band Notifier (email/chat/monitoring protocol).
- Snapshots (every 100 to 250 ms) are published via shared memory/IPC for dashboards.
- A WAL (write-ahead log) provides recovery.

Throughput and latency targets.

- Per update: $O(1)$ arithmetic on a few cache-resident arrays, with no dynamic allocation.
- Pin threads, isolate cores, use huge pages, and apply NUMA-aware placement.
- Batch roughly
$$N \approx 256$$
records per read to amortize overhead.
- For backpressure, rings expose occupancy; when near full, throttle feeders or drop to a degraded aggregate-only mode.

Reference prototype (Python, hot path). Array-based state; easily JIT-compiled with numba or ported to C++ for production.

```
import numpy as np
from dataclasses import dataclass

@dataclass
class Limits:
    q_max: np.ndarray
    notl_max: np.ndarray
    dv01_max: np.ndarray
    vega_max: np.ndarray
    # shape (M,)

class RealTimeRisk:
    def __init__(self, M: int, dv01_per_share, vega_per_share):
        self.M = M

        # state arrays
        self.q = np.zeros(M, dtype=np.float64)       # position
        self.price = np.zeros(M, dtype=np.float64)   # last price
        self.notl = np.zeros(M, dtype=np.float64)    # notional

        # linear Greek per share
        self.dv01_sh = np.asarray(dv01_per_share, dtype=np.float64)
        self.vega_sh = np.asarray(vega_per_share, dtype=np.float64)
        self.dv01 = np.zeros(M, dtype=np.float64)
        self.vega = np.zeros(M, dtype=np.float64)

        # limits (set later)
        self.lim = None

    def set_limits(self, lim: Limits):
        self.lim = lim

    # hot-path update: O(1)
    def on_update(self, inst_id: int, dq: float, price: float) -> int:
        i = inst_id
        self.q[i] += dq
        self.price[i] = price
        self.notl[i] = self.q[i] * price
```

```
        self.dv01[i] = self.q[i] * self.dv01_sh[i]
        self.vega[i] = self.q[i] * self.vega_sh[i]
        return self._check(i)

    # returns bitmask of breaches for instrument i
    def _check(self, i: int) -> int:
        if self.lim is None:
            return 0
        mask = 0
        eps = 1e-9
        if abs(self.q[i]) > self.lim.q_max[i] + eps:
            mask |= 1 << 0
        if abs(self.notl[i]) > self.lim.notl_max[i] + eps:
            mask |= 1 << 1
        if abs(self.dv01[i]) > self.lim.dv01_max[i] + eps:
            mask |= 1 << 2
        if abs(self.vega[i]) > self.lim.vega_max[i] + eps:
            mask |= 1 << 3
        return mask
```

Alerting & reliability.

- Each breach produces
$$\langle ts,\ inst_id,\ metric,\ val,\ limit \rangle$$
on an Alert Queue; the notifier applies hysteresis/debounce and routes by desk.
- Recovery: append every applied update to a WAL; on restart, load the last snapshot and replay the WAL to reconstruct state (typically ≤ 1 s).
- Monitoring: export per-stage latency, ring occupancy, and breach rates; auto-escalate when the risk engine lags.

Why it scales to 10^6/s.

- Updates touch only a constant number of cache-resident doubles (positions, two Greeks, and notional).
- Batching and SPSC rings remove locks.
- Preallocated arrays avoid GC/allocation overhead.
- Horizontal scaling is achieved by sharding instruments across cores; global aggregates can be computed by a second-stage reducer at lower frequency.

Core skill tested. This question tests whether you can turn a production-style requirement into a state model with explicit invariants. A strong answer names the entities you store, the updates you apply on each event, and the correctness conditions you refuse to violate even under latency pressure.

How to present it in interview. In a live interview, open with the data model, then walk through the event path in order: ingest, normalize, update, publish, and recover from gaps or failures. Say the words correctness, sequencing, and recovery explicitly; that is usually what differentiates a trading-systems answer from a toy coding answer.

Common traps.

- Giving a fast design without specifying what state is authoritative under reorders, cancels, or sequence gaps.
- Talking only about happy-path latency and ignoring recovery, backpressure, and stale-state publication.
- Assuming the sketch is production-safe without naming invariants, clocking, or correctness checks.

Related follow-ups.

- What changes if messages arrive out of order or you need to recover from a dropped sequence range?
- How would you test correctness and latency without relying on a single happy-path benchmark?

13.8.2 Order books, VWAP, and production market-data infrastructure

Question 33 Systems / Order Book

Institution type: Quantitative investment management firm
Role: Quantitative Researcher | Difficulty: Medium

Problem. Implement a limit order book class supporting add, cancel, and match with price-time priority.

Solution.

Use FIFO queues at each price level, heaps for discovering the best bid/ask, and an order-id map that points directly to mutable order objects. Cancellation can then mark an order dead in expected $O(1)$, while matching purges canceled orders from the front before executing.

```
from collections import defaultdict, deque
import heapq

class LOB:
    def __init__(self):
        self.bids = defaultdict(deque)
        self.asks = defaultdict(deque)
        self.bid_heap = []          # negative prices
        self.ask_heap = []
        self.orders = {}            # id -> [id, qty, timestamp, side, price]
        self.clock = 0

    def add(self, oid, side, price, qty):
        if oid in self.orders or qty <= 0:
            raise ValueError("invalid order")
        order = [oid, qty, self.clock, side, price]
        self.clock += 1
        self.orders[oid] = order
        book = self.bids if side == 'B' else self.asks
        book[price].append(order)
        heapq.heappush(self.bid_heap if side == 'B' else self.ask_heap,
                       -price if side == 'B' else price)
        return self.match()
```

```
    def cancel(self, oid):
        order = self.orders.pop(oid, None)
        if order is not None:
            order[1] = 0            # lazy deletion; O(1)

    def _best(self, side):
        book = self.bids if side == 'B' else self.asks
        heap = self.bid_heap if side == 'B' else self.ask_heap
        while heap:
            price = -heap[0] if side == 'B' else heap[0]
            q = book.get(price)
            while q and q[0][1] == 0:
                q.popleft()
            if q:
                return price
            heapq.heappop(heap)
            book.pop(price, None)
        return None

    def match(self):
        fills = []
        while True:
            bid = self._best('B')
            ask = self._best('S')
            if bid is None or ask is None or bid < ask:
                break
            b = self.bids[bid][0]
            a = self.asks[ask][0]
            qty = min(b[1], a[1])
            # The older order is resting; trade at its limit price.
            trade_price = bid if b[2] < a[2] else ask
            fills.append((b[0], a[0], trade_price, qty))
            b[1] -= qty
            a[1] -= qty
            if b[1] == 0:
                self.orders.pop(b[0], None)
                self.bids[bid].popleft()
            if a[1] == 0:
                self.orders.pop(a[0], None)
                self.asks[ask].popleft()
        return fills
```

Price-time priority is enforced by selecting the best price first and then the FIFO order at that price. Heap entries are lazily invalidated. In a production exchange simulator, also define modify semantics, self-trade prevention, tick-size validation, partial cancels, and deterministic sequencing.

Core skill tested. This question tests whether you can turn a production-style requirement into a state model with explicit invariants. A strong answer names the entities you store, the updates you apply on each event, and the correctness conditions you refuse to violate even under latency pressure.

How to present it in interview. In a live interview, open with the data model, then walk through the event path in order: ingest, normalize, update, publish, and recover from gaps or failures. Say the words correctness, sequencing, and recovery explicitly; that is usually what

differentiates a trading-systems answer from a toy coding answer.

Common traps.

- Giving a fast design without specifying what state is authoritative under reorders, cancels, or sequence gaps.
- Talking only about happy-path latency and ignoring recovery, backpressure, and stale-state publication.
- Assuming the sketch is production-safe without naming invariants, clocking, or correctness checks.

Related follow-ups.

- What changes if messages arrive out of order or you need to recover from a dropped sequence range?
- How would you test correctness and latency without relying on a single happy-path benchmark?

13.9 Professional Skills, Governance, and Communication

Use these questions to practice complete solutions, concise explanations, and likely follow-up questions.

13.9.1 Behavioral, communication, and research process

Question 34 Behavioural

Institution type: Global investment bank and financial services firm
Role: Risk / Quant | Difficulty: Easy

Problem. Describe a time you identified a significant risk others missed (STAR format).
Solution.

1. **Situation:** context and why it mattered.
2. **Task:** your responsibility and target outcome.
3. **Action:** data you pulled, checks you ran, stakeholders engaged, and what you built (e.g. stress/monitor).
4. **Result:** quantified impact avoided, changes adopted, and what you learned.

Core skill tested. This is a risk-and-finance framing problem. The best answer defines the object precisely, states the minimal formula or workflow, and then immediately mentions the practical assumption set, because interviewers care about whether you know where textbook simplifications break in a desk setting.

How to present it in interview. Answer these by sequencing definition -> construction -> caveat. For example, define the measure or sensitivity, say how you would compute it on real data or positions, and then mention one limitation such as liquidity, model error, or convention dependence.

Common traps.

- Quoting a definition without specifying horizon, confidence level, or market convention.

- Ignoring model risk, liquidity, or scenario design when moving from textbook to practice.
- Giving a sensitivity or valuation formula without saying what inputs must be bootstrapped or estimated.

Related follow-ups.

- What part of the workflow is convention-dependent, and what part is model-dependent?
- How would you explain the same concept to a portfolio manager instead of a quant interviewer?

13.9.2 Model validation, regulation, and governance

Question 35 Model Validation & Governance

Institution type: Global investment bank and financial services firm
Role: Risk / Quant | Difficulty: Medium

Problem. Describe your approach to model validation and governance.
Solution.

1. **Conceptual soundness:** theory, assumptions, intended use, and model boundaries.
2. **Data quality:** lineage, representativeness, missing values and outliers, point-in-time handling, and survivorship-bias controls.
3. **Implementation verification:** code review, benchmark tests, and numerical-stability checks.
4. **Outcomes analysis:** backtests, performance metrics, stability or PSI, and stress behavior.
5. **Ongoing monitoring:** KPIs, KRIs, drift detection, and periodic recalibration triggers.
6. **Governance:** inventories, documentation, change control, independent review, and periodic re-approval.

Core skill tested. This item tests whether you can move cleanly from definition to estimation to validation. A polished response states the estimator or framework, clarifies its assumptions, and then points out the most likely failure mode such as leakage, non-stationarity, or unstable inference.

How to present it in interview. A concise interview answer should always separate modelling from evaluation. State the model or diagnostic, then immediately describe the validation design and the reason you trust or distrust the result under realistic data conditions.

Common traps.

- Stating an estimator but not the assumptions needed for interpretation.
- Forgetting to separate train/validation/test logic or time ordering.
- Choosing a metric that does not match the business or modelling objective.

Related follow-ups.

- What breaks first if the data are non-stationary, heavy-tailed, or missing not at random?
- How would you redesign the validation scheme if the labels overlap or the data are strictly time ordered?

Cross-Industry Foundation Questions

The final 15 questions reinforce mathematical and financial ideas that recur across institution types. They are deliberately chosen for breadth rather than to reproduce every variation of a familiar puzzle or derivation.

13.10 Brain Teasers

Use these questions to practice complete solutions, concise explanations, and likely follow-up questions.

13.10.1 Problem Simplification

Question 36 Screwy Pirates

Institution type: Cross-industry quantitative-finance employers
Category: Problem Simplification

Problem. Five perfectly rational pirates must divide 100 gold coins. The most senior pirate proposes a split and all pirates vote, including the proposer. A proposal passes with at least 50% of the votes; otherwise the proposer is killed and the process repeats with the next pirate. Preferences are lexicographic: survive first, maximize gold second, and, if otherwise indifferent, prefer fewer surviving pirates. How are the 100 coins divided among five pirates?

Solution.

Answer. The most senior pirate keeps 98 coins and gives one coin each to the two pirates who would receive nothing if he were killed; the remaining two pirates get zero.

Reasoning. Work backward. With two pirates, the senior pirate can approve his own proposal. With three pirates, the new senior pirate buys the vote of the pirate who would get zero in the two-pirate outcome for one coin. The same logic alternates as pirates are added. In the five-pirate case, the senior pirate needs three votes including his own, so he buys the two cheapest votes for one coin each and keeps the remaining 98.

Core skill tested. This problem tests structural reasoning: identify the invariant, symmetry, backward-induction rule, or information constraint before doing arithmetic.

How to present it in interview. State the key structural observation first, reduce to a smaller or symmetric case if possible, then finish with a short proof or construction.

Common traps.

- Brute-forcing cases before identifying the invariant.
- Assuming symmetry that the rules do not actually provide.
- Giving a construction without proving that it is minimal or guaranteed.

Related follow-ups.

- Can you generalize the argument to n objects or a different threshold?
- Is there a second solution using an invariant, parity, or information-theoretic bound?

13.10.2 Logic Reasoning

Question 37 Defective Ball

Institution type: Cross-industry quantitative-finance employers
Category: Logic Reasoning

Problem. Among 12 visually identical balls, exactly one has a different weight and may be either heavier or lighter. Using only a balance scale, identify the odd ball and whether it is heavy or light in at most three weighings.

Solution.

Label the balls 1–12. First weigh $1, 2, 3, 4$ against $5, 6, 7, 8$.

If they balance, the odd ball is among 9–12. Weigh $9, 10, 11$ against three known-normal balls. If that balances, 12 is odd and a final comparison with a normal ball identifies whether it is heavy or light. If it does not balance, the direction tells whether 9–11 are heavy or light; compare two of them to identify the ball.

If the first weighing is left-heavy, the candidates are $1, 2, 3, 4$ heavy or $5, 6, 7, 8$ light. Weigh $1, 2, 5$ against $3, 6, 9$, where 9 is known normal. A left-heavy result leaves $1_H, 2_H, 6_L$; a right-heavy result leaves $3_H, 5_L$; a balance leaves $4_H, 7_L, 8_L$. One final comparison resolves each case. The right-heavy first-weighing case is symmetric. Thus three weighings always suffice.

Core skill tested. This problem tests structural reasoning: identify the invariant, symmetry, backward-induction rule, or information constraint before doing arithmetic.

How to present it in interview. State the key structural observation first, reduce to a smaller or symmetric case if possible, then finish with a short proof or construction.

Common traps.

- Brute-forcing cases before identifying the invariant.
- Assuming symmetry that the rules do not actually provide.
- Giving a construction without proving that it is minimal or guaranteed.

Related follow-ups.

- Can you generalize the argument to n objects or a different threshold?
- Is there a second solution using an invariant, parity, or information-theoretic bound?

13.11 Calculus and Linear Algebra

Use these questions to practice complete solutions, concise explanations, and likely follow-up questions.

13.11.1 Important Calculus Methods

Question 38 Root-Finding Algorithms

Institution type: Cross-industry quantitative-finance employers
Category: Important Calculus Methods

Problem. Explain several numerical root-finding algorithms for solving f(x)=0, including their convergence tradeoffs.

Solution.

Three standard choices are:

- **Bisection:** maintain a bracket $[a, b]$ with opposite signs and halve it each step. It is robust and converges linearly.

- **Newton:** $x_{n+1} = x_n - f(x_n)/f'(x_n)$. Near a simple root it converges quadratically, but it needs derivatives and a reasonable starting point.
- **Secant:** replace the derivative by a two-point slope. It needs no analytic derivative and has superlinear convergence, but is less robust than bisection.

In production, a safeguarded hybrid such as Brent's method is often preferable.

Core skill tested. This problem tests whether foundational calculus or linear algebra can be turned into a compact derivation rather than recalled as an isolated formula.

How to present it in interview. Write the governing identity or transformation first, keep notation clean, and name the numerical-stability or regularity caveat when relevant.

Common traps.

- Losing a domain or regularity condition.
- Using a numerically fragile algebraic formula when a factorization is available.
- Skipping the check that the derived solution satisfies the original equation.

Related follow-ups.

- How would you verify the result numerically?
- What changes in higher dimensions or when the matrix or function is poorly conditioned?

13.11.2 Linear Algebra

Question 39 Least Squares via QR

Institution type: Cross-industry quantitative-finance employers
Category: Linear Algebra

Problem. If your programming environment has no built-in linear least-squares routine, how would you implement least squares using matrix factorizations?

Solution.

Given a design matrix X and response vector y, compute a thin QR decomposition

$$X = QR,$$

where $Q^TQ = I$ and R is upper triangular. Least squares minimizes $\|X\beta - y\|_2$, which is equivalent to solving

$$R\hat{\beta} = Q^Ty.$$

Use back substitution for $\hat{\beta}$. This is numerically preferable to explicitly forming $(X^TX)^{-1}X^Ty$, because the normal equations square the condition number and amplify numerical error.

Core skill tested. This problem tests whether foundational calculus or linear algebra can be turned into a compact derivation rather than recalled as an isolated formula.

How to present it in interview. Write the governing identity or transformation first, keep notation clean, and name the numerical-stability or regularity caveat when relevant.

Common traps.

- Losing a domain or regularity condition.
- Using a numerically fragile algebraic formula when a factorization is available.

- Skipping the check that the derived solution satisfies the original equation.

Related follow-ups.

- How would you verify the result numerically?
- What changes in higher dimensions or when the matrix or function is poorly conditioned?

13.12 Probability Theory

Use these questions to practice complete solutions, concise explanations, and likely follow-up questions.

13.12.1 Basic Probability Definitions and Set Operations

Question 40 Coin Toss Game

Institution type: Cross-industry quantitative-finance employers
Category: Basic Probability Definitions and Set Operations

Problem. Gambler A flips n+1 fair coins and gambler B independently flips n fair coins. What is the probability that A obtains strictly more heads than B?

Solution.

The probability is $\boxed{1/2}$. Couple A's first n coins with B's n coins. By symmetry, A is ahead after those n paired coins as often as B is ahead. If the paired totals tie, A's extra coin breaks the tie in A's favor with probability 1/2. A direct binomial calculation gives the same result: for independent $X \sim \mathrm{Bin}(n+1, 1/2)$ and $Y \sim \mathrm{Bin}(n, 1/2)$, $P(X > Y) = 1/2$.

Core skill tested. This problem tests probability modeling: define the sample space or random variables correctly, then use conditioning, symmetry, indicators, or order statistics efficiently.

How to present it in interview. Say what is random and what is conditioned on before calculating. Prefer a one-line symmetry or indicator argument over long enumeration when possible.

Common traps.

- Mixing ordered and unordered outcomes.
- Conditioning on an event without renormalizing.
- Treating dependent events as independent or ignoring a sampling protocol.

Related follow-ups.

- Can you derive the same result with indicators or a complement?
- How does the answer change under biased or dependent sampling?

13.12.2 Conditional Probability and Bayes' Formula

Question 41 Double-Headed Coin Posterior

Institution type: Cross-industry quantitative-finance employers
Category: Conditional Probability and Bayes' Formula

Problem. Among 1,000 coins, one is double-headed and the other 999 are fair. Choose one coin uniformly at random and observe 10 consecutive heads. What is the posterior probability that you chose the double-headed coin?

Solution.

Let U be the double-headed coin and H^{10} the event of ten heads. Bayes' rule gives

$$P(U \mid H^{10}) = \frac{(1)(1/1000)}{(1)(1/1000) + (2^{-10})(999/1000)} = \boxed{\frac{1024}{2023}} \approx 0.5062.$$

Core skill tested. This problem tests probability modeling: define the sample space or random variables correctly, then use conditioning, symmetry, indicators, or order statistics efficiently.

How to present it in interview. Say what is random and what is conditioned on before calculating. Prefer a one-line symmetry or indicator argument over long enumeration when possible.

Common traps.

- Mixing ordered and unordered outcomes.
- Conditioning on an event without renormalizing.
- Treating dependent events as independent or ignoring a sampling protocol.

Related follow-ups.

- Can you derive the same result with indicators or a complement?
- How does the answer change under biased or dependent sampling?

Question 42 Monty Hall

Institution type: Cross-industry quantitative-finance employers
Category: Conditional Probability and Bayes' Formula

Problem. In the standard Monty Hall game, one of three doors hides a car and the other two hide goats. After you choose a door, the host—who knows the prizes—opens a different door showing a goat and offers you the chance to switch to the remaining closed door. Should you switch, and what is your probability of winning if you do?

Solution.

Switch. Your initial door is correct with probability 1/3 and wrong with probability 2/3. Monty always reveals a goat among the other doors. If your initial choice was wrong, the only remaining unopened door is the car; if it was right, switching loses. Therefore

$$\boxed{P(\text{win by switching}) = \frac{2}{3}}.$$

Core skill tested. This problem tests probability modeling: define the sample space or random variables correctly, then use conditioning, symmetry, indicators, or order statistics efficiently.

How to present it in interview. Say what is random and what is conditioned on before calculating. Prefer a one-line symmetry or indicator argument over long enumeration when possible.

Common traps.

- Mixing ordered and unordered outcomes.
- Conditioning on an event without renormalizing.
- Treating dependent events as independent or ignoring a sampling protocol.

Related follow-ups.

- Can you derive the same result with indicators or a complement?
- How does the answer change under biased or dependent sampling?

13.12.3 Expected Value, Variance & Covariance

Question 43 Coupon Collection

Institution type: Cross-industry quantitative-finance employers
Category: Expected Value, Variance & Covariance

Problem. There are N equally likely coupon types, independently drawn. (A) What is the expected number of draws needed to collect all N types? (B) After n draws, what is the expected number of distinct types collected?

Solution.

Suppose there are N equally likely coupon types.

A. Time to complete the set. After $i-1$ distinct types have been collected, the probability the next coupon is new is $(N-i+1)/N$, so the expected waiting time for the next new type is $N/(N-i+1)$. Therefore

$$\boxed{E[T] = N \sum_{k=1}^{N} \frac{1}{k} = NH_N}.$$

B. Distinct types after n draws. Let I_j indicate that type j has appeared. Then

$$P(I_j = 1) = 1 - \left(1 - \frac{1}{N}\right)^n.$$

By linearity of expectation,

$$\boxed{E[D_n] = N\left[1 - \left(1 - \frac{1}{N}\right)^n\right]}.$$

Core skill tested. This problem tests probability modeling: define the sample space or random variables correctly, then use conditioning, symmetry, indicators, or order statistics efficiently.

How to present it in interview. Say what is random and what is conditioned on before calculating. Prefer a one-line symmetry or indicator argument over long enumeration when possible.

Common traps.

- Mixing ordered and unordered outcomes.
- Conditioning on an event without renormalizing.
- Treating dependent events as independent or ignoring a sampling protocol.

Related follow-ups.

- Can you derive the same result with indicators or a complement?
- How does the answer change under biased or dependent sampling?

13.13 Stochastic Process and Stochastic Calculus

Use these questions to practice complete solutions, concise explanations, and likely follow-up questions.

13.13.1 Markov Chain

Question 44 Gambler's Ruin – Unequal Win Probability

Institution type: Cross-industry quantitative-finance employers
Category: Markov Chain

Problem. Player M starts with \$1 and player N with \$2. In each game, the winner receives \$1 from the loser. M wins each game independently with probability 2/3. Play continues until one player is bankrupt. What is the probability that M wins all \$3?

Solution.

Let p_i be the probability that M eventually owns all \$3 when M currently owns i dollars. Then $p_0 = 0$, $p_3 = 1$, and because M wins each game with probability 2/3,

$$p_1 = \frac{2}{3}p_2, \qquad p_2 = \frac{2}{3} + \frac{1}{3}p_1.$$

Solving,

$$p_1 = \frac{2}{3}\left(\frac{2}{3} + \frac{1}{3}p_1\right) = \frac{4}{9} + \frac{2}{9}p_1,$$

so

$$\boxed{p_1 = \frac{4}{7}}.$$

This is the gambler's-ruin absorption probability starting from M's initial wealth of \$1.

Core skill tested. This problem tests stochastic-process state design, stopping-time reasoning, or the use of martingales/Markov structure to avoid unnecessary path enumeration.

How to present it in interview. Define the state or stopping time, write the recursion/PDE/martingale, then state boundary conditions before solving.

Common traps.

- Choosing a state that forgets information needed for the future.
- Applying optional stopping without checking the setup.
- Confusing an unconditional mean-zero process with a martingale.

Related follow-ups.

- Can you solve it both with a Markov recursion and a martingale argument?
- What changes with drift, bias, or different absorbing boundaries?

13.13.2 Dynamic Programming

Question 45 Three-Roll Optimal Stopping

Institution type: Cross-industry quantitative-finance employers
Category: Dynamic Programming

Problem. You may roll a fair six-sided die at most three times. After the first or second roll, you may either accept that roll's face value in dollars and stop, or discard it and continue. The third roll, if reached, must be accepted. What is the optimal strategy and fair value of the game?

Solution.

Work backward.

On the third roll there is no choice, so its value is

$$V_3 = E[X] = 3.5.$$

After the second roll, keep $4, 5, 6$ and reroll $1, 2, 3$. Therefore

$$V_2 = \frac{3(3.5) + 4 + 5 + 6}{6} = \frac{25.5}{6} = 4.25.$$

After the first roll, keep 5 or 6, and reroll $1, 2, 3, 4$, because continuation is worth 4.25. Hence

$$V_1 = \frac{4(4.25) + 5 + 6}{6} = \boxed{\frac{14}{3}} \approx 4.667.$$

Optimal policy: after roll 1 stop on 5–6; after roll 2 stop on 4–6; on roll 3 accept whatever appears.

Core skill tested. This problem tests stochastic-process state design, stopping-time reasoning, or the use of martingales/Markov structure to avoid unnecessary path enumeration.

How to present it in interview. Define the state or stopping time, write the recursion/PDE/martingale, then state boundary conditions before solving.

Common traps.

- Choosing a state that forgets information needed for the future.
- Applying optional stopping without checking the setup.
- Confusing an unconditional mean-zero process with a martingale.

Related follow-ups.

- Can you solve it both with a Markov recursion and a martingale argument?
- What changes with drift, bias, or different absorbing boundaries?

13.14 Finance

Use these questions to practice complete solutions, concise explanations, and likely follow-up questions.

13.14.1 Option Pricing

Question 46 Put–Call Parity

Institution type: Cross-industry quantitative-finance employers
Category: Option Pricing

Problem. State and prove put-call parity for European options on a non-dividend-paying stock.

Solution.

For a non-dividend-paying stock, compare two portfolios with maturity T:

- one European call plus cash worth K at maturity;
- one European put plus one share of stock.

Their terminal payoffs are identical:

$$(S_T - K)^+ + K = (K - S_T)^+ + S_T = \max(S_T, K).$$

No-arbitrage therefore implies

$$\boxed{C + Ke^{-rT} = P + S_0},$$

or

$$\boxed{C - P = S_0 - Ke^{-rT}}.$$

With a continuous dividend yield q, replace S_0 by $S_0 e^{-qT}$.

Core skill tested. This problem tests no-arbitrage reasoning, replication, risk-neutral valuation, or disciplined interpretation of option/rate sensitivities.

How to present it in interview. Start from payoff replication or the pricing measure, write the minimal formula, and immediately state the assumptions under which the sign or equality holds.

Related original-bank record. `a related question`.

Common traps.

- Quoting a Greek or formula without its carry/dividend assumptions.
- Treating a local hedge as a risk-free arbitrage.
- Ignoring convexity, tail risk, units, or discounting conventions.

Related follow-ups.

- Can you derive the result from both replication and risk-neutral valuation?
- Which assumption is most likely to fail on a real desk?

13.14.2 The Greeks

Question 47 Dynamic Delta Hedging

Institution type: Cross-industry quantitative-finance employers
Category: The Greeks

Problem. You are long a European call and want to delta-hedge it with the underlying stock. What initial stock position should you take, and how should you rebalance after a sudden increase in the stock price?

Solution.

A long call has positive delta. To make the position locally delta-neutral, short

$$\boxed{\Delta_C \text{ shares per call}}$$

of stock (with the appropriate contract multiplier).

A vanilla long call has positive gamma, so if GM rises, the call delta increases and the existing short-stock hedge is no longer large enough; short additional shares. If GM falls, delta decreases; buy back some of the short stock.

This is dynamic delta hedging. In continuous Black–Scholes it removes infinitesimal spot risk, but in practice discrete rebalancing leaves gamma, jump, transaction-cost, and model risk.

Core skill tested. This problem tests no-arbitrage reasoning, replication, risk-neutral valuation, or disciplined interpretation of option/rate sensitivities.

How to present it in interview. Start from payoff replication or the pricing measure, write the minimal formula, and immediately state the assumptions under which the sign or equality holds.

Common traps.

- Quoting a Greek or formula without its carry/dividend assumptions.
- Treating a local hedge as a risk-free arbitrage.
- Ignoring convexity, tail risk, units, or discounting conventions.

Related follow-ups.

- Can you derive the result from both replication and risk-neutral valuation?
- Which assumption is most likely to fail on a real desk?

13.14.3 Other Finance Questions

Question 48 Value at Risk

Institution type: Cross-industry quantitative-finance employers
Category: Other Finance Questions

Problem. Define Value at Risk (VaR). What are important shortcomings of VaR, especially for nonlinear derivative portfolios?

Solution.

Value at Risk at confidence level α and horizon h is a loss quantile: the threshold v such that losses exceed v with probability about $1-\alpha$. In notation for a loss variable L,

$$\mathrm{VaR}_\alpha(L) = \inf\{\ell : P(L \le \ell) \ge \alpha\}.$$

Major limitations are especially important for derivatives:

- VaR says little about how large losses are beyond the quantile;
- nonlinear payoffs and volatility/skew make normal approximations unreliable;
- estimates are sensitive to model, horizon, and historical window;
- VaR is not subadditive in full generality, so it can reward apparent diversification incorrectly.

Expected shortfall is often preferred for tail severity because it averages losses beyond VaR and is coherent under standard definitions.

Core skill tested. This problem tests no-arbitrage reasoning, replication, risk-neutral valuation, or disciplined interpretation of option/rate sensitivities.

How to present it in interview. Start from payoff replication or the pricing measure, write the minimal formula, and immediately state the assumptions under which the sign or equality holds.

Common traps.

- Quoting a Greek or formula without its carry/dividend assumptions.
- Treating a local hedge as a risk-free arbitrage.

- Ignoring convexity, tail risk, units, or discounting conventions.

Related follow-ups.

- Can you derive the result from both replication and risk-neutral valuation?
- Which assumption is most likely to fail on a real desk?

13.15 Algorithms and Numerical Methods

Use these questions to practice complete solutions, concise explanations, and likely follow-up questions.

13.15.1 Algorithms

Question 49 Fisher–Yates Shuffle

Institution type: Cross-industry quantitative-finance employers
Category: Algorithms

Problem. Given access to a uniform random-number generator, shuffle a 52-card deck so that every permutation is equally likely. Give the algorithm and justify its correctness.

Solution.

Use the Fisher–Yates shuffle. For $i = 51, 50, \ldots, 1$, draw

$$j \sim \mathrm{Unif}\{0, 1, \ldots, i\}$$

and swap positions i and j.

At step i, each of the $i+1$ remaining cards is equally likely to be placed in position i. Multiplying the conditional probabilities shows every one of the 52! permutations has probability

$$\boxed{1/52!}.$$

The runtime is $O(52)$, or $O(n)$ for n items.

Core skill tested. This problem tests algorithm design: identify the invariant/data structure, prove correctness, and meet the stated time/space target.

How to present it in interview. Give the invariant and complexity before code. Then show the update/search logic and mention edge cases such as empty input, overflow, or unknown bounds.

Common traps.

- Giving code without a correctness invariant.
- Hiding an O(n) operation inside an apparently constant-time loop.
- Ignoring overflow, boundary conditions, or worst-case behavior.

Related follow-ups.

- Can the solution be made streaming or in-place?
- What are the worst-case time and memory bounds, and how would you test them?

13.15.2 Numerical Methods

Question 50 Monte Carlo Variance Reduction

Institution type: Cross-industry quantitative-finance employers
Category: Numerical Methods

Problem. Explain several variance-reduction techniques that can improve the efficiency of Monte Carlo pricing.

Solution.

Common variance-reduction techniques include:

- **Antithetic variates:** pair a path driven by Z with one driven by $-Z$; averaging helps when the payoffs are negatively correlated.
- **Control variates:** use a correlated quantity Y with known expectation and estimate $X - \beta(Y - E[Y])$, choosing β near the regression-optimal coefficient.
- **Importance sampling:** simulate from a distribution that visits the payoff-relevant region more often and reweight by the likelihood ratio; especially useful for rare events/deep OTM payoffs.
- **Stratified or quasi-Monte Carlo sampling:** cover the input space more evenly than independent pseudorandom draws.
- **Moment matching:** adjust a simulated sample so low-order moments better match the target distribution, though this should be used carefully to avoid hidden bias.

The best method depends on the payoff and on which source of variance dominates.

> **Core skill tested.** This problem tests algorithm design: identify the invariant/data structure, prove correctness, and meet the stated time/space target.
>
> **How to present it in interview.** Give the invariant and complexity before code. Then show the update/search logic and mention edge cases such as empty input, overflow, or unknown bounds.
>
> **Common traps.**
>
> - Giving code without a correctness invariant.
> - Hiding an O(n) operation inside an apparently constant-time loop.
> - Ignoring overflow, boundary conditions, or worst-case behavior.
>
> **Related follow-ups.**
>
> - Can the solution be made streaming or in-place?
> - What are the worst-case time and memory bounds, and how would you test them?